PART I – Multiple choice questions

Instructions for multiple choice questions: For each question there is *at least* one correct answer alternative. In other words, for each question the number of correct alternatives is 1, 2, 3, or 4. Mark your answers on the separate answer sheet, *not* on the exam. You are awarded one point for each question if, and only if, you have selected the correct alternatives.

1. What is the relationship between *scientific methods* and *methodology*? Select all true claims.

   a) Methodology is about method choice.
   b) Methodology provides empirical data for problems that methods cannot solve.
   c) Methodology is the process of constructing specific methods.
   d) Methodology is another word for a collection of methods.

2. What is true about *lexical* definitions? Select all true claims.

   a) A lexical definition attempts to capture the common usage of a term.
   b) A lexical definition can be imprecise.
   c) A lexical definition stipulates how a word ought to be used.
   d) A lexical definition is intended to capture how the author intends to use the term.

3. What is *realism* in the theory of science? Select all true claims. Realism is...

   a) ...the view that we have reason to believe that our best theories of the world are literally true.
   b) ...the view that scientific concepts are not descriptions of the actual world, but merely convenient fictions with which scientists can construct coherent theories.
   c) ...the view that there is an existing reality independent of us and our senses.
   d) ...the view that scientific theories describe, or aim at describing, what the world is really like.

4. What is true about *inductive inferences*? Select all true claims. Inductive inferences ...

   a) ...require a justification.
   b) ...do not guarantee that the conclusions are true.
   c) ...have premises.
   d) ...are fallible.

5. What is true of Karl Popper's Falsificationism? Select all correct answers. The falsificationist...

   a) ...never accepts any hypothesis as true – only maintains non-falsified hypotheses as so far not rejected.
   b) ...conjectures (proposes) falsifiable hypotheses.
   c) ...increases her confidence in the truth of a hypothesis if this hypothesis is falisfiable.
   d) ...rejects the hypothetico-deductive model.

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6. Which of the following sentences express good hypotheses?

   a) "A square is a geometric shape with four sides."
   b) "Do not add water to strong acids!"
   c) "Is Amyloid plaque the cause of Alzheimer's?"
   d) "Computer program of type A uses more memory than than computer program of type B."

7. What are *necessary* steps in a measurement procedure? Select all correct alternatives.

   a) Representation by a measurement scale.
   b) Specification of a measure and definition of a unit.
   c) Specification of measurement operation(s).
   d) Establishing the naturalness of the measure.

8. What are *necessary* conditions for a good operationalization? Select all correct alternatives.

   a) The operationalisation must be theory-independent.
   b) The observable property must be observable in practice with sufficient precision.
   c) The property of interest must be defined with reference to the operationalization.
   d) The connection between property of interest and observable property must be sufficiently stable for practical purposes.

9. Which of the following claims about systematic error is correct? Select all true claims.

   a) Experimental blinding is a strategy to reduce systematic error.
   b) Increasing construct validity is not a viable strategy to reduce systematic error.
   c) Systematic error can be reduced by repeating measurements and averaging over outcomes.
   d) Systematic errors are (in principle) predictable and and deviations from the true value are typically constant or proportional to the true value.

10. What are necessary aspects of scientific experiments? Select all correct alternatives.

    a) The intervention.
    b) The laboratory.
    c) The statistical model.
    d) The manipulation of some real variable.

11. What is correct about blinding in experiments? Select all correct alternatives.

    a) Blinding can only be realized through randomization.
    b) One condition for single blinding is that subjects do not know which treatment they are getting.
    c) One condition for double blinding is that experimenters do not know which treatments are administered to what subjects.
    d) Single blinding is an effective strategy to reduce placebo effects and thus to increase the internal validity of an experiment.

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12. What is correct about the *repeatability* of experiments? Select all correct alternatives.

a) The requirement of repeatability does not mean that scientific experiments actually have to be repeated, only that it should be possible to repeat them given the published description.
b) An experiment is repeatable if competent persons, different from the experimenters, who follow the given description of the experiment, can repeat it.
c) Repeatability does not imply replication.
d) It is sufficient for an experiment to be repeatable that one of the participating experimenters knows how to repeat the experiment at a later stage.

13. Identify the relevant epistemic virtues of scientific models from the following list.

a) Purpose independence
b) Simplicity
c) Non-computability
d) Transparency

14. Why should hypotheses be tested statistically? Select all correct answers.

a) Observations might contain random measurement errors. A single (seemingly) falsifying observation might thus not be a good reason to reject a hypothesis.
b) Some hypotheses only make claims about distributions of outcomes.
c) Neither confirmation nor falsification can be based on deductive relationships of hypotheses and observations.
d) Using probability calculus is a way to model and calculate the degree of confirmation that some evidence confers to a hypothesis.

15. Which claims are true about Bayesian statistics? Select all correct alternatives.

a) Bayesian statistics can evaluate a potentially large set of competing hypotheses.
b) In Bayesian statistics, the prior probability of a hypothesis is changed when new evidence is observed.
c) The main goal in Bayesian hypothesis evaluation is to decide whether to accept a single hypothesis as either true or false.
d) Bayesian statistics offers an unambiguous algorithm for assigning prior probabilities to hypotheses.

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16. Which claims are true about the relation between correlation and causation? Select all correct claims.

    a) Correlation between two variables is symmetric. The causal relation between two variables in asymmetric.
    b) That there is a common cause of two variables X and Y explains why there is a causal relation between X and Y (either X causes Y or Y causes X), although there is no correlation between X and Y.
    c) Many different causal models between two variables are compatible with the observation that the two variables are correlated.
    d) Correlation measures the association between variables. Causation measures the productive influence of one variable on another.

17. What is true of tacit knowledge?

    a) Teaching tacit knowledge requires extensive personal contact, regular interaction and trust
    b) The ability to ride a bicycle is an example of tacit knowledge
    c) To know something tacitly means that one has the ability to describe this knowledge intuitively, i.e. without first having to study it
    d) Tacit knowledge is taught mainly through engineering textbooks and manuals

18. Which of the following strategies can help improve the accuracy of interpretation of behavior in a usability experiment? Select all correct alternatives.

    a) Use inter-subjective checks.
    b) Make ad hoc modifications whenever suitable.
    c) Avoid interpretation of mental states (like beliefs or desires).
    d) Check for coherence by repeating questions in questionnaire or interview.

19. What is true about *informed consent*? Select all correct alternatives. Informed consent ...

    a) ... only has support among deontologists who think that individual autonomy is morally important.
    b) ... is the faculty dean's approval of the initiation of a new study.
    c) ... constitutes the legitimate authorization of research participation.
    d) ... is usually understood as having three conditions that need to be fulfilled.

20. A decision has to be made between four alternative courses of action. On alternative A, there are two possible outcomes: 100 people dying (probability 0.6), or no one dying (probability 0.4). On alternative B, there are four possible outcomes (equal probabilities): 20 people dying, 100 people dying, 400 people dying or 4 people dying. On C, there are only two possible outcomes: 800 people dying (probability 0.01) or no one dies (probability 0.99) On D, finally, either 30 people die, or no one dies (equal probabilities). Select all correct statement(s), given the expected value criterion for decisions involving risks:

    a) B is the fourth best alternative
    b) A is the best alternative
    c) C is the best alternative
    d) A is the third best alternative

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TaMoS: Answer sheet for multiple choice questions

Student's name: **Correct answers, NS version August 21, 2018**

Personal number: Natural versions

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PART I – Multiple choice questions

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1. What is true about knowledge according to the classical definition? Select all correct alternatives. According to the classical definition of knowledge...

   a) ...if you know that it is raining then you are justified in believing that it is raining.
   b) ...if you know that it is raining at place P at time t, then you believe with certainty that it is raining at place P at time t.
   c) ...if you know that it is raining at place P at time t, then it is raining at place P at time t.
   d) ...you cannot know that it is raining at place P at time t.

2. What is correct about the peer review system? Select all true claims.

   a) It is a system created to uphold scientific quality of the submitted paper, especially regarding the novelty and justification of its claims.
   b) Peer reviewers recommend either acceptance, various forms of revision, or rejection of the manuscript submitted to a journal.
   c) Editors across scientific journals cooperate so that bad articles sent to their respective journals will not be accepted.
   d) For any experimental or empirical study, the peer review includes a repetition of the experiment or study.

3. What is correct about the *hypothetico-deductive* method for hypothesis testing? Select all true claims.

   a) If the outcome of a test (i.e. an experiment) is negative, this does not necessarily mean that the hypothesis should be rejected. The negative result might be explained by an auxiliary hypothesis being false.
   b) It helps deducing hypotheses from empiricial data.
   c) Hypotheses are tested through observable consequences.
   d) The method determines how hypotheses can be deduced from other, more fundamental theories.

4. What distinguishes deductive inferences from inductive inferences? Select all true claims.

   a) Projection is an example of inductive inference, *modus ponens* an example of deductive inferences.
   b) Generalisation is an example of an inductive inference, *modus tollens* is an example of deductive inferences.
   c) Deductions might be true in the right circumstances, inductions are never true.
   d) Modus tollens is an example of an inductive inference, projection is an example of a deductive inference.

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5. What is true of the Quine-Duhem thesis? Select all true claims. According to the Quine-Duhem thesis ...

   a) ... inductive inference are never valid
   b) ... we never test a single hypothesis alone.
   c) ... auxiliary hypotheses can be neglected
   d) ... falsification is clearly preferable to induction

6. Which of the following claims about systematic error is correct? Select all true claims.

   a) Systematic errors can sometimes be reduced by experimental blinding.
   b) Increasing construct validity is not a viable strategy to reduce systematic error.
   c) Systematic errors are (in principle) predictable.
   d) Systematic error can be reduced by repeating measurements and averaging over outcomes.

7. What are *necessary* conditions for a good operationalization? Select all correct alternatives.

   a) The property of interest must be defined with reference to the operationalization.
   b) The connection between property of interest and observable property must be valid according to our best knowledge.
   c) The observable property must be observable in practice with sufficient precision.
   d) It must be applied exclusively to properties that are directly observable only with aids.

8. What is correct about blinding in experiments? Select all correct alternatives.

   a) One condition for double blinding is that experimenters do not know which treatments are administered to what subjects.
   b) Single blinding is an effective strategy to reduce placebo effects and thus to increase the internal validity of an experiment.
   c) Experimental blinding is *not* an instance of experimental control by elimination.
   d) Double blinding implies that the experimenter is ignorant of which variables are dependent and which are independent.

9. Which advantages do experiments have over other empirical methods for testing hypotheses?

   a) Experiments, unlike other empirical methods, also offer *confirming* evidence for a hypothesis.
   b) Experiments offer greater control of many relevant background conditions than other empirical methods.
   c) With experiments, the Duhem–Quine problem does not arise.
   d) Causal hypotheses are more easily tested with experiments.

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10. Which of the following claims about disturbing factors in experiments is correct? Select all true alternatives.

    a) Experiments with disturbing factors always allow externally valid inferences.
    b) Randomization is an important but not necessary tool for dealing with disturbing factors.
    c) The best way to control disturbing factors in an experiment is through statistical techniques.
    d) Disturbing factors are only present in experiments on humans and animals.

11. What is true about valid isolating models? Select all correct claims.

    a) Although of theoretical interest, isolating models play no relevant role for any of the engineering sciences.
    b) Isolating models can never be validated by experiments.
    c) For an isolating model to be valid, the components or aspects it isolates must be truly independent from the target's other components or aspects.
    d) A valid isolating model is dissimilar to its target and still licenses relevant (externally) valid inferences.

12. What is the relation between models and experiments? Select all correct claims.

    a) Experiments and models are similar in that they always are mere representations of targets.
    b) Internal validity is generally less of a problem for models than for experiments.
    c) Both models and experiments involve manipulations. But in models, these are always manipulations of representations.
    d) The problems posed by flawed proofs and programming bugs in models is generally a greater threat to internal validity than the problem of experimental control poses for experiments.

13. Which claims are true of Fisher's Null Hypothesis Testing? Select all correct options.

    a) Fisher's Null Hypothesis Test determines the probability of the null hypothesis, given the data.
    b) How the significance level is set is a mere convention.
    c) Fisher's Null Hypothesis Test only helps decide whether to reject the null hypothesis or not.
    d) The p-value measures whether the data was produced by chance or not

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14. Consider the following descriptions of three treatments:
(A) "A medical intervention results in a 33% relative decrease in the incidence of fatal myocardial infarction".
(B) "A medical intervention results in a 2.6 %-point decrease in the incidence of fatal myocardial infarction – 2.7% vs. 5.3%".
(C) "91 persons must be treated with the intervention to reduce the number of fatal myocardial infarction from 3 to 2".
What is true of these descriptions? Select all correct options.

a) (A) represents an absolute risk reduction.
b) It has been shown that most people consider treatments represented in format (A) more effective than treatments represented in format (C).
c) It has been shown that most people consider treatments represented in format (C) more effective than treatments represented in format (A).
d) Any intervention correctly described by (C) is also correctly described by (A).

15. What is true of the Deductive-Nomological (DN) Account of Explanation? Select all true answers.

a) According to the DN account, we achieve genuine understanding of a phenomenon by identifying laws that give reasons to expect the phenomenon to occur.
b) The DN account is problematic, because some cases satisfy the DN account's conditions, but intuitively do not constitute genuine explanations.
c) According to the DN account, the logical structure of scientific explanations and predictions is very similar.
d) The DN-account cannot use mathematical equations in its explanans.

16. What is true of tacit knowledge?

a) Teaching tacit knowledge requires extensive personal contact, regular interaction and trust.
b) Tacit knowledge is difficult to transfer to another person by means of writing it down or verbalizing it.
c) Tacit knowledge is taught mainly through engineering textbooks and manuals.
d) The ability to ride a bicycle is an example of tacit knowledge.

17. Which claims are true either of methodological individualism or holism? Select all true claims.

a) Methodological individualism claims that there are no social entities in the world, only individuals (and families).
b) Methodological holism defends the view that parts of a whole are in intimate interconnection, such that they cannot exist independently of the whole.
c) Methodological holism is compatible with the claim that all social phenomena are produced by individual interactions.
d) Methodological holism claims that sometimes, individualist accounts do not provide illuminating explanations of social phenomena because these phenomena can be instantiated in many different ways on the individual level.

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18. Which of these are theories of normative ethics? Select all correct answers.

a) Deontology.
b) Informed consent.
c) Virtue ethics.
d) Consequentialism.

19. What is true about the *precautionary principle*? Select all correct answers. The precautionary principle ...

a) ... implies that the scientist is responsible for showing that e.g. a new detergent is not harmful, or that the new species does not harm the ecosystem, etc.
b) ... is of no concern in ideal democracies.
c) ... imposes a moral constraint on researchers.
d) ... states that the public bears the burden of proof to show that a substance is harmful.

20. What is true about how risks are perceived? Select all correct alternatives. Psychological studies have shown that ...

a) ... new risks are seen as worse than old risks.
b) ... high probability for smaller negative consequences is seen as worse than small probabilities for a very bad consequence.
c) ... risk to identifiable individuals are seen as worse than statistical risks (with same or greater number of expected deaths).
d) ... natural risks are seen as worse than technological risks.

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TaMoS: Answer sheet for multiple choice questions

Student's name: **Correct answers, NS and CS Versions Dec 20, 2017**

Personal number: Natural and computational versions

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PART I – Multiple choice questions

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1. What is the relationship between *scientific methods* and *methodology*? Select all true claims.

   a) Methodology is another word for a collection of methods.
   b) Methodology is the process of constructing specific methods.
   c) For different methods, the same methodology might apply.
   d) Methodology provides empirical data for problems that methods cannot solve.

2. Does methodology require scientists to completely eliminate the risk of error in science? Select all correct answers.

   a) No. One argument for this is that science cannot eliminate all errors, only manage potential sources of error.
   b) No. Errors are a necessary part of accurate claims.
   c) No. Almost all empirical claims are fallible, so an empirical science that aims to completely eliminate error would lose much of its content.
   d) Yes. Erroneous claims are unscientific.

3. What distinguishes deductive inferences from inductive inferences? Select all true claims.

   a) Valid deduction is truth-preserving, justified induction is fallible.
   b) While justifications of inductive inference rules are controversial, deductive inference rules are "natural" in the sense that they do not require any justification.
   c) Projection is an example of inductive inference, *modus ponens* an example of deductive inferences.
   d) Modus tollens is an example of an inductive inference, projection is an example of a deductive inference.

4. What is true about *falsifiable statements*? Select all true claims. A falsifiable statement...

   a) ...is not true.
   b) ...may be false.
   c) ...is always a claim about a particular instance.
   d) ...is testable even though it may be about something not directly observable.

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5. What is true of Karl Popper's falsificationism? Select all correct answers. The falsificationist...

   a) ...rejects the hypothetico-deductive model.
   b) ...increases her confidence in the truth of a hypothesis if this hypothesis is falisfiable.
   c) ...seeks to falsify hypotheses through confronting them with observable evidence.
   d) ...conjectures (proposes) falsifiable hypotheses.

6. Which of the following is true of interval scales? Select all true claims.

   a) Zero on the scale represents the complete absence of the property concerned.
   b) On (linear) interval scales, arithmetic means are meaningful as averages.
   c) If a quantity is measured on an interval scale, it cannot also be measured on a ratio scale.
   d) 30 degrees Celsius is twice as warm as 15 degrees Celsius.

7. What are *necessary* steps in a measurement procedure? Select all correct alternatives.

   a) Specification of the measure in terms of a physical constant.
   b) Definition of the concept to be measured.
   c) Specification of measurement operation(s).
   d) Specification of a measure and definition of a unit.

8. What is correct about the *repeatability* of experiments? Select all correct alternatives.

   a) The requirement of repeatability does not mean that scientific experiments actually have to be repeated, only that it should be possible to repeat them given the published description.
   b) An experiment is repeatable if there is enough information available about the experiment so that a repetition of the experiment based on this information will yield the same result.
   c) If an experiment is repeatable, then it is also reproducible.
   d) Repeatability of an experiment can be determined by analysing the published material about this experiment.

9. Which factors would threaten the *internal validity* of inferences from an experiment? Select all correct alternatives.

   a) Failing to justify why an experimental result applies to a relevant real-world situation.
   b) Failing to control background conditions.
   c) Misunderstanding the influence of experimental manipulation.
   d) Failing to accurately identify the relevant background conditions.

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10. Which of the following claims about disturbing factors in experiments is correct? Select all true alternatives.
    a) The best way to control disturbing factors in an experiment is through statistical techniques.
    b) Randomization is an important but not necessary tool for dealing with disturbing factors: by eliminating selection bias and also facilitating blinding.
    c) Observer influence may be a source of disturbing factors in an experiment.
    d) Disturbing factors are only present in experiments on humans and animals.

11. What are good (epistemic or non-epistemic) reasons why scientists build and investigate models of targets they are interested in, rather than investigating these targets directly? Select all correct alternatives.
    a) Some targets are too complex for it to be possible to gain knowledge from directly investigating them.
    b) Targets can never be manipulated directly.
    c) Some targets cannot be handled directly for practical reasons.
    d) We often do not know anything about the target.

12. What is the relation between models and experiments? Select all correct claims.
    a) Internal validity is generally less of a problem for models than for experiments.
    b) Experiments and models are similar in that they always are mere representations of targets.
    c) Both models and experiments involve manipulations. But in models, these are always manipulations of representations.
    d) External validity is more of a problem for experiments than for models.

13. Which of the following claims about statistics are true? Select all correct options.
    a) One can only "lie with statistics" by using invalid statistical methods, or by applying valid statistical methods to junk data.
    b) Statistics provides a toolbox of descriptive and inferential methods; and statistical reasoning justifies the choice from this toolbox.
    c) It is always clear from the descriptions of the statistical concepts and tools themselves when one should apply them.
    d) Statistics always lies.

14. Which claims are true about Bayesian statistics? Select all correct alternatives.
    a) The main goal in Bayesian hypothesis evaluation is to decide whether to accept a single hypothesis as either true or false.
    b) In Bayesian statistics, the probability of a hypothesis is changed if new evidence is observed.
    c) In Bayesian statistics, no probabilities are assigned to hypotheses.
    d) Bayesian statistics interprets probabilities as "degrees of belief" or "subjective confidence in hypotheses".

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15. What is true of the Deductive-Nomological (DN) Account of Explanation? Select all true answers.

a) The DN account is problematic, because some cases satisfy the DN account's conditions, but intuitively do not constitute genuine explanations.
b) According to the DN account, an explanation consists in the inductive inference from initial conditions to the explanandum.
c) According to the DN account, the logical structure of scientific explanations and predictions is very similar.
d) According to the DN account, we achieve genuine understanding of a phenomenon by identifying laws that give reasons to expect the phenomenon to occur.

16. In what ways does qualitative and quantitative data differ? Select all correct alternatives.

a) Quantitative approaches require operationalisations; qualitative methods do not.
b) Qualitative and quantitative data share no information in common – they are fundamentally different kinds of data.
c) Large qualitative data sets are more difficult to summarize than large quantitative data sets.
d) To produce quantitative data (measured on interval or ratio scales), a specification of a unit for comparison is required; this is not needed for the production of qualitative data.

17. In the social sciences, how can the quality of an interpretation of human action be improved? Select all true claims.

a) Only rely on mental states that are directly observable.
b) Generate additional evidence for the assumed cognitive processes, e.g. through eye tracking or thinking aloud protocols.
c) Check for coherence between answers, e.g. through (slightly reformulated) repeat questions.
d) Use intersubjective checks of the interpretation, e.g. by involving multiple researchers in the analysis.

18. Which of these are theories of normative ethics? Select all correct answers.

a) Consequentialism.
b) Virtue ethics.
c) Moral justification.
d) Prohibition of plagiarism.

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19. What is true regarding the steps taken in the methodology of design?

a) When deciding between possibilities, you can either aim for optimization (doing an exhaustive search) or satisficing (determining a minimal threshold).
b) The methodology of design consists of the same steps as the methodology of natural science.
c) The first step is to perform an exhaustive search for a complete set of design alternatives.
d) The design needs to be validated, which means to show that it actually fulfils the functional requirements.

20. Which of these are technical definitions of risk? Select all correct alternatives. Risk is...

a) ...the statistical expected value of unwanted events which may or may not occur.
b) ...the severity of an unwanted event which may or may not occur.
c) ...an unwanted event which has already occurred.
d) ...the probability of an event occurring or not occurring.

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TaMoS: Answer sheet for multiple choice questions

Student's name: **Correct answers, NS version December 17 2018**

Personal number: Natural versions

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1. What is true about *episteme* and *techne*? Select all correct alternatives.
   a) Techne consists in knowing why, episteme consists in knowing how.
   b) Techne provides the ability of achieving some practical goals.
   c) Episteme means theoretical knowledge, techne means practical knowledge.
   d) Techne has deep and complex methodological problems of its own.

2. Which of the following claims about words and definitions is/are true? Select all correct alternatives.
   a) In a lexical definition of a value-laden term, the term's value-ladenness should be accurately reflected.
   b) If a definiens applies to fewer objects than its associated definiendum, then the definition is too broad.
   c) If a definition is too broad, then it cannot be too narrow.
   d) If a definition contains vague words, then it is necessarily too narrow.

3. What is true about *falsifiable statements*? Select all true claims. Falsifiable statements ...
   a) ... are necessarily false.
   b) ... are always claims about particular instances.
   c) ... is testable but may not be directly observable.
   d) ... are not true.

4. Which of the following is observable only indirectly? Select all correct alternatives.
   a) The charge in a battery.
   b) The well-being of a child.
   c) The pH value of a liquid.
   d) The precise temperature of a liquid.

5. Which of the following expresses a value on a *ratio scale*?
   a) In garments-sizes, M is smaller than L.
   b) Water freezes at a temperature of 0 degrees Celsius.
   c) Usain Bolt set a world record, running 100 m in 9.58 seconds.
   d) A ship travels at 20 nautical miles per hour.

6. What are necessary steps in a measurement procedure? Select all correct alternatives.
   a) Specification of the measure in terms of a physical constant.
   b) Establishing the naturalness of the measure.
   c) Specification of a measure and definition of a unit.
   d) Representation by a measurement scale.

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7. Which of the following claims about experiments and observational studies are correct? Select all true claims.

   a) In observational studies one manipulates only the variables that are thought to influence the outcome.
   b) Experiments include control and manipulation.
   c) Inferences from observational studies might be as internally valid as from experiments.
   d) In an observational study one has limited control over the relevant variables, while in an experiment one always has absolute control.

8. What is correct about the *repeatability* of experiments? Select all correct alternatives.

   a) It is sufficient for an experiment to repeatable that one of the participating experimenters knows how to repeat the experiment at a later stage.
   b) An experiment is repeatable if other competent persons that follow the given description of the experiment can repeat it.
   c) Repeatability does not imply replicatability.
   d) If an experiment is repeatable, then it is also reproducible.

9. Identify the relevant epistemic virtues of scientific models from the following list.

   a) Generality
   b) Theory-independence
   c) Simplicity
   d) Non-computability

10. What is the relation between models and experiments? Select all correct claims

    a) External validity is more of a problem for experiments than for models.
    b) Models are always representations of targets different from themselves, while experiments are sometimes performed on the target systems themselves.
    c) Experiments and models are similar in that they always are mere representations of targets.
    d) Internal validity is generally less of a problem for models than for experiments.

11. Which of the following claims about statistics are true? Select all correct options.

    a) It is always clear from the descriptions of the statistical concepts and tools themselves when one should apply them.
    b) One can only "lie with statistics" by using invalid statistical methods, or by applying valid statistical methods to junk data.
    c) Statistics offers different algorithms that might yield different results when applied to the same kind of data.
    d) Statistics always lies.

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12. Which claims are true about Bayesian statistics? Select all correct alternatives.

    a) Bayesian statistics offers an unambiguous algorithm for assigning prior probabilities to hypotheses.
    b) In Bayesian statistics, no probabilities are assigned to hypotheses.
    c) The "problem of old evidence" remains a substantial problem for Bayesian statistics.
    d) The main goal in Bayesian hypothesis evaluation is to decide whether to accept a single hypothesis as either true or false.

13. Which claims are true about the relation between correlation and causation? Select all correct claims.

    a) Many different causal models between two variables are compatible with the observation that the two variables are correlated.
    b) Mill's method of difference has been largely ignored by the modern practice of randomised controlled trials.
    c) That there is a common cause of two variables X and Y explains why there is a causal relation between X and Y (either X causes Y or Y causes X), although there is no correlation between X and Y.
    d) By analysing correlation data, we typically are able to determine which causal model is the unique explanation of the data.

14. Which of the following are *type* explanations? Select all correct alternatives.

    a) These vases broke because they were dropped.
    b) The cars crashed because the road went from dual to single lane while one of the cars was trying to overtake.
    c) Birds are able to follow their migration routes because they have receptors in their eyes that are sensitive to the magnetic field of the earth.
    d) Roads that repeatedly go from double to single file and back again have higher accident rates because drivers are bad at judging overtaking maneuvers.

15. What are the main claims of anti-naturalist views of the social sciences? Select all correct alternatives.

    a) The social sciences must employ their own methods, different from the natural sciences - but both the natural and the social sciences can use the same methodology.
    b) The social sciences, in contrast to the natural sciences, focus on individual cases and cannot generalise from them.
    c) Although the social sciences use qualitative data, in contrast to the natural sciences, there are methods by which this data can be quantified and thus be analyzed with natural science methods.
    d) The social sciences, in contrast to the natural sciences, are in the business of interpreting human behaviour. Interpretation does not produce causal explanations, but some other form of understanding.

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16. Which claims are true either of methodological individualism or holism? Select all true claims.

a) Methodological individualism claims that all good social explanations provide individualist accounts of the explanandum.
b) Methodological holism claims that sometimes, individualist accounts do not provide illuminating explanations of social phenomena because they can be instantiated in many different ways on the individual level.
c) Methodological holism is compatible with the claim that all social phenomena are produced by individual interactions.
d) Methodological individualism claims that there are no social entities in the world, only individuals (and families).

17. Scientific openness is generally valued in science, however there are legitimate exceptions. Select all correct alternatives. Not sharing scientific data *may* be justified when ...

a) ... sharing the data jeopardizes the chances of publishing the results in an article.
b) ... protecting one's institution from legitimate moral blame.
c) ... protecting people's privacy.
d) ... the correctness of the data has not been checked.

18. What is true about the *precautionary principle*? Select all correct answers. The precautionary principle ...

a) ... states that the proponent of an activity, rather than the public, should bear the burden of proof of the activity's harmlessness.
b) ... implies that the scientist must be able to show that the new detergent is not harmful, or that the new species does not harm the ecosystem, etc.
c) ... imposes a moral constraint on researchers.
d) ... is of no concern in ideal democracies.

19. What is true about risk assessment? Select all correct alternatives. Risk assessment is the step in risk analysis where ...

a) ... evidential information is communicated to the public.
b) ... policies are put in place.
c) ... regulations are determined.
d) ... experiments or observational studies are performed.

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20. A decision has to be made between four alternative courses of action.
On alternative A1, there are three possible outcomes: 20 people dying (probability 0.2), 500 people dying (p = 0.002), or 5 people dying (p = 0.798).
On alternative A2, there are only two possible outcomes (identical probabilities): 20 people dying, or no one dying.
On A3, there are three possible outcomes (identical probabilities): 9 people dying, 4 people dying, or 12 people dying.
On A4, finally, there is a one percent probability of 800 people dying (otherwise, no one dies).
Select all correct statement(s), given the expected value criterion for decisions involving risks:

a) A2 is the fourth best alternative.
b) A3 is the second best alternative.
c) A3 is the best alternative.
d) A1 is the worst alternative.

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TaMoS: Answer sheet for multiple choice questions

Student's name: **Correct answers, January 11, 2017, nat. sci. version**

Personal number:

	a)	b)	c)	d)
1.		×	×	×
2.	×			
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5.			×	×
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	a)	b)	c)	d)
11.			×	
12.			×	
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PART I – Multiple choice questions

Instructions for multiple choice questions: For each question there is *at least* one correct answer alternative. In other words, for each question the number of correct alternatives is 1, 2, 3, or 4. Mark your answers on the separate answer sheet, *not* on the exam. You are awarded one point for each question if, and only if, you have selected the correct alternatives.

1. What is *realism* in the theory of science? Select all true claims. Realism is...

   a) ...the view that there is an existing reality independent of us and our senses.
   b) ...the view that scientific progress is impossible.
   c) ...the view that a good theory is a theory that is instrumentally valuable.
   d) ...the view that scientific theories describe, or aim at describing, what the world is really like.

2. What is the relationship between *scientific methods* and *methodology*? Select all true claims.

   a) Methodology is another word for a collection of methods.
   b) Methods tell you how to do things, methodology justifies why you do things this way rather than that way.
   c) Methodology provides empirical data for problems that methods cannot solve.
   d) For different methods, the same methodology might apply.

3. Which of the following sentences express good hypotheses?

   a) "Computer program of type A uses more memory than than computer program of type B."
   b) "The Ebola epidemic was caused by bacteria."
   c) "Individuals with high level of education judge climate change as a high risk."
   d) "Is Amyloid plaque the cause of Alzheimer's?"

4. What is an ad hoc hypothesis? Select all correct alternatives.

   a) It is a modification of a theory that does not propose a new means of testing the theory.
   b) It is a modification of a theory that logically contradicts the rest of the theory.
   c) It is a modification of a theory that decreases falsifiability
   d) It is a kind of claim that shares some (or all) observable consequences with the previous hypothesis.

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5. What is true of Karl Popper's falsificationism? Select all correct answers. The falsificationist...

   a) ...considers only those hypotheses as true that have withstood many falsification attempts.
   b) ...never accepts any hypothesis as true – only maintains non-falsified hypotheses as so far not rejected.
   c) ...seeks to falsify hypotheses with observable evidence.
   d) ...conjectures (proposes) falsifiable hypotheses.

6. What is true of the 7 base units of the International System of Units (SI)?

   a) Some of the units may be more fundamental than the others.
   b) It is generally accepted that the SI base units constitute the natural units.
   c) An unlimited number of derived units can be formed from the SI base units through mathematical operations.
   d) It consists of the units: ampere, kelvin, mole, candela, meter, kilogram and second.

7. What are *necessary* steps in a measurement procedure? Select all correct alternatives.

   a) Specification of a measure and definition of a unit.
   b) Specification of measurement operation(s).
   c) Establishing the naturalness of the measure.
   d) Specification of the measure in terms of a physical constant.

8. What is correct about the *repeatability* of experiments? Select all correct alternatives.

   a) It is sufficient for an experiment to be repeatable that one of the participating experimenters knows how to repeat the experiment at a later stage.
   b) If an experiment is repeatable, then it is also reproducible.
   c) The criterion of repeatability is satisfied when a person reports that she has done a number of trials with different experimental setups for each trial and out of these trials succeeded in reproducing the same result twice.
   d) The requirement of repeatability does not mean that scientific experiments actually have to be repeated, only that it should be possible to repeat them given the published description.

9. Which of the following claims about disturbing factors in experiments is correct? Select all true alternatives.

   a) Observer influence may be a source of disturbing factors in an experiment.
   b) Disturbing factors are only present in experiments on humans and animals.
   c) One eliminates a disturbing factor by making sure that the disturbing factor is not present.
   d) When we cannot eliminate disturbing factors, we try to separate them and register their influence independently

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10. Which advantages do experiments have over other empirical methods for testing hypotheses?

a) Causal hypotheses are more easily tested with experiments.
b) With experiments, the Duhem–Quine problem does not arise.
c) Experiments, unlike other empirical methods, also offer *confirming* evidence for a hypothesis.
d) Experiments offer greater control of many relevant background conditions than other empirical methods.

11. What is the relation between models and experiments? Select all correct claims.

a) Performing experimental control is often a difficult task. Setting variable and parameter values in mathematical models, in contrast, is often trivial.
b) Both models and experiments involve manipulations. But in models, these are always manipulations of representations.
c) Internal validity is generally less of a problem for models than for experiments.
d) The problems posed by flawed proofs and programming bugs in models is generally a greater threat to internal validity than the problem of experimental control poses for experiments.

12. What are good (epistemic or non-epistemic) reasons why scientists build and investigate models of targets they are interested in, rather than investigating these targets directly? Select all correct alternatives.

a) Ethical or political reasons prohibit the direct manipulation of some targets.
b) Science is interested in representations, not in the targets themselves.
c) Targets can never be manipulated directly.
d) We often do not know anything about the target.

13. Which claims are true of Fisher's Null Hypothesis Testing? Select all correct options.

a) Fisher's Null Hypothesis Test determines the probability of the null hypothesis, given the data.
b) How the significance level is set is a mere convention.
c) Fisher's Null Hypothesis Test only helps decide whether to reject the null hypothesis or not.
d) The p-value measures whether the data was produced by chance or not.

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14. Consider the following descriptions of three treatments:
(A) "A medical intervention results in a 33% relative decrease in the incidence of fatal myocardial infarction".
(B) "A medical intervention results in a 2.6 %-point decrease in the incidence of fatal myocardial infarction – 2.7% vs. 5.3%".
(C) "91 persons must be treated with the intervention to reduce the number of fatal myocardial infarction from 3 to 2".
What is true of these descriptions? Select all correct options.

a) Any intervention correctly described by (C) is also correctly described by (A).
b) (C) represents the risk reduction in natural frequencies.
c) It has been shown that most people consider treatments represented in format (A) more effective than treatments represented in format (C).
d) (A) represents an absolute risk reduction.

15. What is true about explanans (plural: explanantia) and explanandum (plural: explananda)? Select all true claims.

a) Explanantia that are not fully accurate (and hence false in that sense) can nevertheless sometimes explain.
b) Failing to specify the contrast of an explanandum might lead to multiple correct competing explanations.
c) There are singular and general explananda, but there are only general explanantia.
d) The explanans is what requires an explanation, the explanandum is what provides the explanation.

16. What is true of tacit knowledge?

a) Tacit knowledge is difficult to transfer to another person by means of writing it down or verbalizing it.
b) To know something tacitly means that one has a skill, experience or idea without necessarily being able to describe it explicitly.
c) Being able to fluently apply the Navier–Stokes equations to complex flow phenomena is an example of tacit knowledge.
d) To know something tacitly means that one has the ability to describe something from intuition, i.e. without first having had to study it.

17. What are the main claims of anti-naturalist views of the social sciences? Select all correct alternatives.

a) The social sciences, in contrast to the natural sciences, focus on individual cases and cannot generalise from them.
b) Although the social sciences use qualitative data, in contrast to the natural sciences, there are methods by which these data can be quantified and thus be analyzed with natural science methods.
c) The social sciences derive law-like generalizations from case studies and qualitative data.
d) The social sciences, in contrast to the natural sciences, are in the business of interpreting human behaviour. Interpretation does not produce causal explanations, but some other form of understanding.

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18. What is true about *informed consent*? Select all correct alternatives. Informed consent ...

    a) ... should never be sought if it risks spoiling the study.
    b) ... is the faculty dean's approval of the initiation of a new study.
    c) ... is usually understood as having three conditions that need to be fulfilled.
    d) ... cannot be given involuntarily.

19. Which of these are theories of normative ethics? Select all correct answers.

    a) Consequentialism.
    b) Deontology.
    c) Prohibition of plagiarism.
    d) Informed consent.

20. What is true about how risks are perceived? Select all correct alternatives. Psychological studies have shown that ...

    a) ... high probability for smaller negative consequences is seen as worse than small probabilities for a very bad consequence.
    b) ... natural risks are seen as worse than technological risks.
    c) ... involuntary risks are seen as worse than voluntary risks.
    d) ... risks where one lacks control are seen as worse.

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TaMoS: Answer sheet for multiple choice questions

Student's name: **Correct answers, NS/CS/AF Versions Jan 10, 2018**

Personal number: AK2030/AK2036/AK2050/F1N5113/FAK3024/AK2034/AK2040/DA2205/AF2023

	a)	b)	c)	d)
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	a)	b)	c)	d)
11.	◯	◯	◯	
12.	◯			
13.		◯	◯	
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20.			◯	◯

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PART I – Multiple choice questions

Instructions for multiple choice questions: For each question there is *at least* one correct answer alternative. In other words, for each question the number of correct alternatives is 1, 2, 3, or 4. Mark your answers on the separate answer sheet, *not* on the exam. You are awarded one point for each question if, and only if, you have selected the correct alternatives.

1. What is true about knowledge according to the classical definition? Select all correct alternatives. According to the classical definition of knowledge...

   a) ...if it is not raining at place P at time t then you cannot know that it is raining at place P at time t.
   b) ...if you know that it is raining then you are justified in believing that it is raining.
   c) ...you cannot know that it is raining at place P at time t.
   d) ...if you know that it is raining at place P at time t, then it is raining at place P at time t.

2. What is the relationship between *scientific methods* and *methodology*? Select all true claims.

   a) Methods tell you how to do things, methodology asks why you do things this way rather than that way.
   b) Methodology provides empirical data for problems that methods cannot solve.
   c) Methodology is another word for a collection of methods.
   d) Methodology critically assesses the application of methods for certain purposes.

3. Mark all of the following sentences that express a good hypothesis.

   a) "A square is a geometric shape with four sides."
   b) "Do not add water to strong acids!"
   c) "The Ebola epidemic was caused by bacteria."
   d) "Is Amyloid plaque the cause of Alzheimer's?"

4. What is true of Karl Popper's falsificationism? Select all correct answers. The falsificationist...

   a) ...increases her confidence in a hypothesis if this hypothesis is falisfiable
   b) ...rejects the hypothetico-deductive model
   c) ...aims to reject (but not accept) hypotheses.
   d) ...considers hypotheses as true only if they have withstood many falsification attempts

5. Which of the following is observable only indirectly? Select all correct alternatives.

   a) The exact pH value of a liquid.
   b) The stars outside of our solar system.
   c) The shape of cells.
   d) The well-being of a child.

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6. Which of the following claims about random error is correct? Select all true claims.

   a) Random errors can be reduced by trying an alternative operationalisation.
   b) Lowering the amount of random errors will increase the amount of systematic errors.
   c) Random errors are inherently unpredictable, and have null expected value.
   d) Reducing random errors increases precision, but not accuracy.

7. What is correct about blinding in experiments? Select all correct alternatives.

   a) One condition for single blinding is that subjects do not know which treatment they are getting.
   b) Double blinding implies that the experimenter is ignorant of which variables are dependent and which are independent.
   c) Blinding can only be realized through randomization.
   d) Experimental blinding is *not* an instance of experimental control by elimination.

8. What are necessary aspects of scientific experiments? Select all correct alternatives.

   a) The manipulation of some real variable.
   b) A theoretical hypothesis.
   c) The control of background variables.
   d) The observation of the result of manipulation.

9. Identify the relevant epistemic virtues of scientific models from the following list.

   a) Non-computability
   b) Transparency
   c) Precision
   d) Simplicity

10. What is the relation between models and experiments? Select all correct claims.

    a) Performing experimental control is often a difficult task. Setting variable and parameter values in mathematical models is, in contrast, often trivial.
    b) External validity is more of a problem for experiments than for models.
    c) Models are always representations of targets different from themselves, while experiments are sometimes performed on the target systems themselves.
    d) The problems posed by flawed proofs and programming bugs in models is generally a greater threat to internal validity than the problem of experimental control poses for experiments.

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11. Which of the following claims about statistics are true? Select all correct options.

    a) "Lying with statistics" often consists in the intentional or accidental misapplication of valid statistical methods to good-quality data.
    b) The application of valid statistical tools to good-quality data can lead to false conclusions.
    c) One can only "lie with statistics" by using invalid statistical methods, or by applying valid statistical methods to junk data.
    d) Statistics always lie.

12. Consider the following descriptions of three treatments:
    (A) "A medical intervention results in a 33% relative decrease in the incidence of fatal myocardial infarction".
    (B) "A medical intervention results in a 2.6 %-point decrease in the incidence of fatal myocardial infarction – 2.7% vs. 5.3%".
    (C) "91 persons must be treated with the intervention to reduce the number of fatal myocardial infarction from 3 to 2".
    What is true of these descriptions? Select all correct options.

    a) (A) represents an absolute risk reduction.
    b) Any intervention correctly described by (C) is also correctly described by (A).
    c) It has been shown that most people consider treatments represented in format (C) more effective than treatments represented in format (A).
    d) It has been shown that most people consider treatments represented in format (A) more effective than treatments represented in format (C).

13. Which claims are true about the relation between correlation and causation? Select all correct claims.

    a) Many different causal models involving two variables are compatible with the observation that the two variables are correlated.
    b) By analysing correlation data, we typically are able to determine which causal model is the unique explanation of the data.
    c) That there is a common cause of two variables X and Y explains why there is a causal relation between X and Y (either X causes Y or Y causes X), although there is no correlation between X and Y.
    d) Correlation between two variables is symmetric. The causal relation between two variables in asymmetric.

14. What is true about explanans (plural: explanantia) and explanandum (plural: explananda)? Select all true claims.

    a) The explanans is what requires an explanation, the explanandum is what provides the explanation.
    b) There are singular and general explananda, but there are only general explanantia.
    c) Explanantia that are not fully accurate (and hence false in that sense) can nevertheless sometimes explain.
    d) A singular explanandum can be explained with a singular or general explanans.

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15. Which of the following strategies can help improve the accuracy of interpretation of behavior in a usability experiment? Select all correct alternatives.

    a) Ensure that the belief-desire attributions are circular.
    b) Ensure that the observer has knowledge of the artefact and is competent in its use.
    c) Gather evidence for cognitive processes.
    d) Check for coherence by repeating questions in questionnaire or interview.

16. What is true about *informed consent*? Select all correct alternatives. Informed consent ...

    a) ... should never be sought if it risks spoiling the study.
    b) ... only has support among deontologists who think that individual autonomy is morally important.
    c) ... is not something that everyone can give.
    d) ... designates the legitimate authorization of research participation.

17. A decision has to be made between four alternative courses of action.
    On alternative A1, there are three possible outcomes: 20 people dying (probability 0.2), 500 people dying (p = 0.002), or 5 people dying (p = 0.798).
    On alternative A2, there are only two possible outcomes (equal probabilities): 20 people dying, or no one dying.
    On A3, there are three possible outcomes (equal probabilities): 9 people dying, 4 people dying, or 12 people dying.
    On A4, finally, there is a one percent probability of 800 people dying (otherwise, no one dies).
    Select all correct statement(s), given the expected value criterion for decisions involving risks:

    a) A3 is the second best alternative.
    b) A3 is the best alternative.
    c) A1 is the worst alternative.
    d) A4 is the best alternative.

18. Which of the following claims about words and definitions is/are true? Select all correct alternatives.

    a) In a lexical definition of a value-laden term, the term's value-ladenness should be accurately reflected in the definiens.
    b) If a definiens applies to fewer objects than its associated definiendum, then the definition is too broad.
    c) A vague word lacks a sharp limit between what it applies to and what it does not apply to.
    d) If a definition contains vague words, then it is necessarily too narrow.

19. What is true about *falsifiable statements*? Select all true claims. A falsifiable statement...

    a) ... is testable even though it may be about something not directly observable.
    b) ... is either true or false.
    c) ... is not true.
    d) ... is necessarily false.

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20. Which of the following expresses a value on a *ratio scale*?

a) The distance between the sun and the earth is on average eight light-minutes.
b) Water freezes at a temperature of 0 degrees Celsius.
c) A ship travels at 20 nautical miles per hour.
d) In garments-sizes, M is smaller than L.

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TaMoS: Answer sheet for multiple choice questions

Student's name: **Correct answers, Natural Science June 10, 2017**

Personal number: NATURAL SCIENCE

	a)	b)	c)	d)
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	a)	b)	c)	d)
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PART I – Multiple choice questions

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1. What is true about the *classical definition of knowledge*? Select all correct alternatives. According to the classical definition of knowledge ...
   a) ... if you know that it is raining then you are justified in believing that it is raining.
   b) ... you know that it is raining at place P at time t only if you have observed that it is raining at place P at time t.
   c) ... if it is not raining at place P at time t then you do not know that it is raining at place P at time t.
   d) ... if you know that it is raining at place P at time t, then it is raining at place P at time t.

2. Does methodology aim to eliminate all errors in science? Select all correct answers.
   a) No. Available evidence often gives us reason to choose one method over the other, but these reasons are often neither complete nor certain.
   b) No. One argument for this is that science cannot eliminate all errors, only manage potential sources of error.
   c) No. Almost all our beliefs are fallible, so it would be impossible to completely eliminate error.
   d) No. Errors are a necessary part of accurate claims.

3. What distinguishes valid deductive inferences from inductive inferences? Select all true claims.
   a) While justifications of inductive inference rules are controversial, deductive inference rules are "natural" in the sense that they do not require any justification.
   b) Valid deduction is truth-preserving, justified induction is fallible.
   c) Generalisation is an example of inductive inferences, *modus tollens* is an example of deductive inferences.
   d) Modus tollens is an example of an inductive inference, projection is an example of a deductive inference.

4. What is correct about the *hypothetico-deductive* method for hypothesis testing? Select all true claims.
   a) Operationalization of not directly observable features of interest is necessary for the hypothetico-deductive method.
   b) Observable consequences are deduced from one's hypotheses and tested empirically.
   c) If the outcome of a test (i.e. an experiment) is negative, this does not necessarily mean that the hypothesis should be rejected. The negative result might be explained by an auxiliary hypothesis being false.
   d) The method determines how hypotheses can be deduced from other, more fundamental theories.

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5. What is true about "Hume's Problem of Induction"-argument? Select all true claims. Hume argued ...

   a) ... that there is no necessary connection between past and future observations.
   b) ... that induction necessarily led to false beliefs.
   c) ... that inductive inference practices in science should be replaced by deductive inference practices.
   d) ... that in our inferences, we rely on inference rules that themselves require justification.

6. What is true about *falsifiable statements*? Select all true claims. Falsifiable statements ...

   a) ... are not true.
   b) ... are neither necessarily true nor necessarily false.
   c) ... may be false.
   d) ... are testable even though they may be about something not directly observable.

7. What kind of measurement scale is employed when one ranks movies on the scale 1,2,3,4,5? Select all correct alternatives.

   a) A ratio scale.
   b) A scale that represents less information than a ratio scale.
   c) An interval scale.
   d) The kind of scale that is also used when describing the final positions in a marathon.

8. Which of the following claims about systematic error is correct? Select all true claims.

   a) Systematic error can be reduced by repeating measurements and averaging over outcomes.
   b) Systematic errors are (in principle) predictable and deviations from the true value are typically constant or proportional to the true value.
   c) If a cause of the systematic error is identified, then it can usually be corrected.
   d) Systematic error affects the degree to which repeated measurements under unchanged conditions show the same results.

9. Which factors would threaten the *internal validity* of inferences from an experiment (in contrast to an observational study)? Select all correct alternatives.

   a) Misinterpreting experimental observations.
   b) Misunderstanding the influence of experimental manipulation.
   c) Failing to (materially) control background conditions.
   d) Misunderstanding the influence of background conditions.

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10. What is correct about the *repeatability* of experiments? Select all correct alternatives.

    a) The criterion of repeatability is sufficiently satisfied when a person reports that she has done a number of trials with different experimental setups for each trial and out of these trials succeeded in reproducing the same result twice.
    b) Repeatability of an experiment can be determined by analysing the published material about this experiment.
    c) If an experiment is repeatable, then it is also reproducible.
    d) An experiment is repeatable if other competent persons that follow the given description of the experiment can repeat it.

11. What are good (epistemic or non-epistemic) reasons why scientists build and investigate models of targets they are interested in, rather than investigating these targets directly? Select all correct alternatives.

    a) Targets can never be manipulated directly.
    b) It is too expensive to handle some targets directly.
    c) Science is interested in representations, not in the targets themselves.
    d) Some targets are too complex to gain knowledge from their direct investigation.

12. What is true about valid isolating models? Select all correct claims.

    a) Although of theoretical interest, isolating models play no relevant role for any of the engineering sciences.
    b) A valid isolating model is dissimilar to its target and still licenses relevant (externally) valid inferences.
    c) A valid isolating model singles out one aspect of the target, ignoring all the others.
    d) A valid isolating model represents the workings of the isolated factor accurately.

13. Why should hypotheses be tested statistically? Select all correct options.

    a) Statistics is the only way how to justify inductive inferences.
    b) Statistical accounts often impose weaker demands on hypothesis testing than (non-statistical) accounts of confirmation and falsification.
    c) Using probability calculus is a way to model and calculate the degree of confirmation that some evidence confers to a hypothesis.
    d) Some hypotheses only make claims about distributions of outcomes.

14. Which claims are true about Bayesian statistics? Select all correct alternatives.

    a) Bayesian statistics can evaluate a potentially large set of competing hypotheses.
    b) The main goal in Bayesian hypothesis evaluation is to decide whether to accept a single hypothesis as either true or false.
    c) Bayesian statistics offers an unambiguous algorithm for assigning prior probabilities to hypotheses.
    d) Bayesian statistics interprets probabilities as "degrees of belief" or "subjective confidence in hypotheses".

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15. What is true of the Deductive-Nomological (DN) Account of Explanation? Select all true answers.

a) The DN account is problematic, because some cases satisfy the DN account's conditions, but intuitively do not constitute genuine explanations.
b) According to the DN account, understanding of a phenomenon is achieved by having reasons for why the phenomenon was expected on a lawful basis.
c) According to the DN account, an explanation consists in the deduction of the explanandum from laws of nature and initial conditions.
d) According to the DN account, the logical structure of scientific explanations and predictions is very similar.

16. Which claims are true about the relation between correlation and causation? Select all correct claims.

a) Correlation measures the covariance between variables. Causation measures the influence of one variable on another.
b) That there is a common cause of two variables X and Y explains why there is a causal relation between X and Y (either X causes Y or Y causes X), although there is no correlation between X and Y.
c) By analysing correlation data, we typically are able to determine which causal model is the unique explanation of the data.
d) Variables can be correlated with each other even though no direct causal relation exists between them.

17. What is true about the *precautionary principle*? Select all correct answers. The precautionary principle ...

a) ... is of no concern in ideal democracies.
b) ... states that the proponent of an activity, rather than the public, should bear the burden of proof of the activity's harmlessness.
c) ... states that when experiments exceeds the research budget, they should not be conducted.
d) ... states that when an activity raises threats of harm to human health or the environment, precautionary measures should be taken even if some cause and effect relationships are not fully established scientifically.

18. What is true about *informed consent*? Select all correct alternatives. Informed consent ...

a) ... is not something that everyone can give.
b) ... only has support among deontologists who think that individual autonomy is morally important.
c) ... designates the legitimate authorization of research participation.
d) ... should never be sought if it risks spoiling the study.

19. What is true about risk assessment? Select all correct alternatives. Risk assessment is the step in risk analysis where ...

a) ... policies are put in place.
b) ... experiments or observational studies are performed.
c) ... evidential information is communicated to the public.
d) ... statistical evidence is considered.

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20. A decision has to be made between four alternative courses of action.
On alternative A1, there are three possible outcomes: 20 people dying (probability 0.2), 500 people dying (p = 0.002), or 5 people dying (p = 0.798).
On alternative A2, there are only two possible outcomes (equal probabilities): 20 people dying, or no one dying.
On A3, there are three possible outcomes (equal probabilities): 9 people dying, 4 people dying, or 12 people dying.
On A4, finally, there is a one percent probability of 800 people dying (otherwise, no one dies).
Select all correct statement(s), given the expected value criterion for decisions involving risks:

a) A2 is the best alternative.
b) A1 is the worst alternative.
c) A4 is the best alternative.
d) A1 is the third best alternative.

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TaMoS: Answer sheet for multiple choice questions

Student's name: **Correct answers, March 13, 2017, nat. sci. version**

Personal number:

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PART I – Multiple choice questions

Instructions for multiple choice questions: For each question there is *at least* one correct answer alternative. In other words, for each question the number of correct alternatives is 1, 2, 3, or 4. Mark your answers on the separate answer sheet, *not* on the exam. You are awarded one point for each question if, and only if, you have selected the correct alternatives.

1. What is true about the justification of method choice?

   a) Scientists are sometimes not justified in their method choices.
   b) Satisfying epistemic goals is not the prime justification of method choice.
   c) The practice of justifying method choice is called methodology.
   d) Scientists should choose the methods which give the best economic results.

2. Which of these can be said to be epistemic goals of science?

   a) Identifying the cause (or causes) which produced an event.
   b) Creating profits.
   c) Producing unjustified beliefs.
   d) Knowing that an artefact satisfies a certain function.

3. What is true about Hume's Problem of Induction? Select all true claims. Hume argued ...

   a) ... that there is no *necessary* connection between past and future observations.
   b) ... that induction *necessarily* leads to false beliefs.
   c) ... that induction cannot be justified by foundational principles.
   d) ... that inductive inference practices in science should be replaced by deductive inference practices.

4. What is true of Karl Popper's falsificationism? Select all correct answers. The falsificationist...

   a) ... increases her confidence in the truth of a hypothesis if this hypothesis is falisfiable.
   b) ... proposes falsifiable hypotheses.
   c) ... never accepts any hypothesis as true – only maintains non-falsified hypotheses as so far not rejected.
   d) ... aims to reject (but not to accept) hypotheses.

5. What is true about *inductive inferences*? Select all true claims. Inductive inferences ...

   a) ... require a justification.
   b) ... never lead to true conclusions.
   c) ... are fallible.
   d) ... are explicating, rather than amplifying.

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6. What are *necessary* conditions for a good operationalization? Select all correct alternatives.

   a) The operationalization must be theory-independent.
   b) The operationalization must connect the feature of interest with an observable property.
   c) The connection between feature of interest and an observable property must be sufficiently stable for practical purposes.
   d) One should only operationalize hypotheses that satisfy the quality criteria for hypotheses.

7. Which of the following claims about systematic error is correct? Select all true claims.

   a) Experimental blinding is a strategy to reduce systematic error.
   b) Systematic error can be reduced by repeating measurements and averaging over outcomes.
   c) Systematic error increases the variation of the results from repeated measurements under unchanged conditions.
   d) Increasing construct validity is not a viable strategy to reduce systematic error.

8. What is true about Mill's method of difference?

   a) According to Mill's method of difference, we can determine causal relations by performing experiments.
   b) It requires successful control and intervention.
   c) By activating a factor in one system and not activating it in another, else identical, system, we can determine the effect of that factor.
   d) By activating the same factor in two separate identical systems we can determine the effect of that factor.

9. What is true about Randomised Control Trials (RCTs)?

   a) It ensures that background factors are equally distributed between test and control in a test.
   b) Stratification can help with distributing known background factors equally between test and control group.
   c) Using RCT implies that you are using double blinding.
   d) Subjects are separated into test and control by a random process.

10. What is true regarding the virtues of models?

    a) When comparing two models, the one which implies the parameter specifications of the other, is more robust with respect to that parameter.
    b) A model is simple if changing one of the assumptions in the model does not change the model result.
    c) If changing one of the assumptions in a model does not change the model result, then the model is robust with respect to this assumption.
    d) When a model user can understand how the model produces the result, then it is epistemically transparent.

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11. What distinguishes models from theories? Select all correct claims.

    a) Theories make general claims, models can only represent particulars.
    b) In those scientific domains where models and theories co-exist (e.g. chemistry), theories typically express general principles, which are applied in the building of models.
    c) Theories are empirically supported, models are not.
    d) The main reason to reject a theory is that it is false. The main reason to reject a model is that it is not useful for a given purpose.

12. Which claims are true of Fisher's Significance Testing? Select all correct options.

    a) How the significance level is set is a mere convention.
    b) Fisher's Significance Test determines the probability of the hypothesis, given the data.
    c) Fisher's Significance Test only helps decide whether to reject the hypothesis or not.
    d) The result of Fisher's Significance Testing can be influenced by changing the experimental design, and by changing the partition of possible outcomes.

13. Why should hypotheses be tested statistically? Select all correct answers.

    a) Statistics is the only way to justify inductive inferences.
    b) Statistics provides tools that turn hypothesis testing into a non-fallible procedure.
    c) Using probability calculus is a way to model and calculate the degree of confirmation that some evidence confers to a hypothesis.
    d) Statistics helps us specify an acceptable error margin for accepting or rejecting a hypothesis.

14. What is true regarding explanatory virtues (what makes explanations powerful)?

    a) *Precision of the explanandum* means that an explanation which uses numbers or other specific information to explain a phenomena is better.
    b) An explanation which as its explanans provides correct descriptions of some contributing causes has high precision.
    c) An explanans should identify those contributing causes that produce the difference asked for in the contrast of the explanandum.
    d) An explanation is more powerful than another, if is less sensitive to background causes.

15. What is true of the Deductive-Nomological (DN) Account of Explanation? Select all true answers.

    a) All causal explanations are DN-explanations.
    b) The DN account is problematic, because some cases satisfy the DN account's conditions, but intuitively do not constitute genuine explanations.
    c) According to the DN account, an explanation consists in the inductive inference from initial conditions to the explanandum.
    d) The DN-account cannot use mathematical equations in its explanans.

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16. What is true of tacit knowledge?

    a) Tacit knowledge is taught mainly through engineering textbooks and manuals.
    b) Tacit knowledge is difficult to transfer to another person by means of writing it down or verbalizing it.
    c) Transferring tacit knowledge typically requires personal contact and interaction.
    d) To know something tacitly means that one has the ability to describe this knowledge intuitively, i.e. without first having to study it.

17. In the social sciences, how can the quality of an interpretation of human action be improved?

    a) Check for coherence between answers, e.g. through (slightly reformulated) repeat questions.
    b) Ensure that the observer/interpreter is competent in the task that the observed subject is performing.
    c) Only rely on mental states that are directly observable.
    d) Use intersubjective checks of the interpretation, e.g. by involving multiple researchers in the analysis.

18. What is true about *utilitarianism*? Select all correct alternatives. According to utilitarians ...

    a) ... actions are morally right or wrong in themselves.
    b) ... moral value should be assigned to consequences rather than to duties or rights.
    c) ... each action can be assessed by a utility level, often interpreted as happiness.
    d) ... individuals have rights, and there are some things we cannot do to others without violating these rights.

19. What is true about the *precautionary principle*? Select all correct answers. The precautionary principle ...

    a) ... implies that if scientist A creates a new detergent, then scientist A must be able to show that the new detergent is not harmful.
    b) ... is of no concern in ideal democracies.
    c) ... states that the proponent of an activity, rather than the public, should bear the burden of proof of the activity's harmlessness.
    d) ... states a list of research projects that should never be initiated.

20. Which of these are reasonable ways to manage uncertainty whose probabilities cannot be quantified?

    a) The principle of sufficient reason - we should assume the current state of knowledge is enough to determine the best option.
    b) We should identify the option with the least bad outcome and choose that option - MaxiMin.
    c) We can identify safety factors, such as predicting the maximal load, or safety barriers working independently of the possible harm.
    d) We should identify the option with the worst outcome and choose that option - MaxiMax.

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TaMoS: Answer sheet for multiple choice questions

Student's name: **Correct answers, NS version March 11 2019**

Personal number: Natural versions

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PART I – Multiple choice questions

Instructions for multiple choice questions: For each question there is *at least* one correct answer alternative. In other words, for each question the number of correct alternatives is 1, 2, 3, or 4. Mark your answers on the separate answer sheet, *not* on the exam. You are awarded one point for each question if, and only if, you have selected the correct alternatives.

1. Which of the following claims about words and definitions is/are true? Select all correct alternatives.

   a) If a definiens applies to fewer objects than its associated definiendum, then the definition is too broad.
   b) If a definition is too broad, then it cannot be too narrow.
   c) A vague word lacks a sharp limit between what it applies to and what it does not apply to.
   d) A stipulative definition is not necessarily narrower than its lexical counterpart.

2. What is true about knowledge according to the classical definition? Select all correct alternatives. According to the classical definition of knowledge...

   a) ...if it is not raining at place P at time t then you cannot know that it is raining at place P at time t.
   b) ...if you know that it is raining then you are justified in believing that it is raining.
   c) ...you cannot know that it is raining at place P at time t.
   d) ...if you know that it is raining at place P at time t, then it is raining at place P at time t.

3. What is true of Karl Popper's falsificationism? Select all correct answers. The falsificationist...

   a) ...conjectures (proposes) falsifiable hypotheses.
   b) ...seeks to falsify hypotheses with observable evidence.
   c) ...rejects the hypothetico-deductive model.
   d) ...accepts only those hypotheses as true that have withstood many falsification attempts.

4. What is true about Hume's Problem of Induction? Select all true claims. Hume argued ...

   a) ...that in our inferences, we rely on inference rules that themselves require justification.
   b) ...that induction necessarily leads to false beliefs.
   c) ...that induction cannot be justified by foundational principles.
   d) ...that there is no *necessary* connection between past and future observations.

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5. Which of the following is true of interval scales? Select all true claims.

   a) If a quantity is measured on an interval scale, it cannot also be measured on a ratio scale.
   b) One can meaningfully speak of how many times greater or lesser the difference between two points on the scale is than the difference between another pair of points.
   c) Time measured by calendar years (such as "2015 A.D." or "500 B.C.") is an interval scale.
   d) 30 degrees Celsius is twice as warm as 15 degrees Celsius.

6. Which of the following claims about random error is correct? Select all true claims.

   a) Random errors are inherently unpredictable, and have null expected value.
   b) Reducing random errors increases precision, but not accuracy.
   c) Random errors can be reduced by trying an alternative operationalisation.
   d) Random errors can be reduced by repeating measurements and averaging over outcomes.

7. What is correct about the *repeatability* of experiments? Select all correct alternatives.

   a) It is sufficient for an experiment to repeatable that one of the participating experimenters knows how to repeat the experiment at a later stage.
   b) Repeatability of an experiment can be determined by analysing the published material about this experiment.
   c) The criterion of repeatability is sufficiently satisfied when a person reports that she has done a number of trials with different experimental setups for each trial and out of these trials succeeded in reproducing the same result twice.
   d) The requirement of repeatability does not mean that scientific experiments actually have to be repeated, only that they should be possible to repeat given the published description.

8. What are necessary aspects of scientific experiments? Select all correct alternatives.

   a) The intervention.
   b) The statistical model.
   c) The control of background variables.
   d) The manipulation of some real variable.

9. Identify the relevant epistemic virtues of scientific models from the following list.

   a) Free from idealizations
   b) Precision
   c) Purpose independence
   d) Simplicity

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10. What are good (epistemic or non-epistemic) reasons why scientists build and investigate models of targets they are interested in, rather than investigating these targets directly? Select all correct alternatives.

a) It is too expensive to handle some targets directly.
b) Ethical or political reasons prohibit the direct manipulation of some targets.
c) Some targets cannot be handled directly for practical reasons.
d) Science is interested in representations, not in the targets themselves.

11. Consider the following descriptions of three treatments:
(A) "A medical intervention results in a 33% relative decrease in the incidence of fatal myocardial infarction".
(B) "A medical intervention results in a 2.6 %-point decrease in the incidence of fatal myocardial infarction – 2.7% vs. 5.3%".
(C) "91 persons must be treated with the intervention to reduce the number of fatal myocardial infarction from 3 to 2".
What is true of these descriptions? Select all correct options.

a) (A) represents an absolute risk reduction.
b) It has been shown that most people consider treatments represented in format (C) more effective than treatments represented in format (A).
c) Any intervention correctly described by (C) is also correctly described by (A).
d) (B) represents an absolute risk reduction.

12. Which of the following claims about statistics are true? Select all correct options.

a) The application of valid statistical tools to good-quality data can lead to false conclusions.
b) Statistics always lies.
c) Statistics offers different algorithms that might yield different results when applied to the same kind of data.
d) One can only "lie with statistics" by using invalid statistical methods, or by applying valid statistical methods to junk data.

13. What is true of the Deductive-Nomological (DN) Account of Explanation? Select all true answers.

a) According to the DN account, an explanation consists in the inductive inference from initial conditions to the explanandum and laws of nature.
b) According to the DN account, an explanation consists in the deduction of the explanandum from true generalizations and initial conditions.
c) According to the DN account, the logical structure of scientific explanations and predictions is very similar.
d) The DN account is problematic, because some cases satisfy the DN account's conditions, but intuitively do not constitute genuine explanations.

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14. Which of the following are *type* explanations? Select all correct alternatives.
    a) These five vases broke because they were dropped.
    b) Birds are able to follow their migration routes because they have receptors in their eyes that are sensitive to the magnetic field of the earth.
    c) Most house fires are due to unattended candles.
    d) The cars crashed because the road went from dual to single lane while one of the cars was trying to overtake.

15. Which of the following strategies can help improve the accuracy of interpretation of behavior in a usability experiment? Select all correct alternatives.
    a) Make ad hoc modifications whenever suitable.
    b) Ensure that the belief-desire attributions are circular.
    c) Ensure that the observer has knowledge of the artefact and is competent in its use.
    d) Gather evidence for cognitive processes.

16. What is true about *deontology*? Deontologists ...
    a) ... are more inclined than others to think in terms of categorical imperatives.
    b) ... think that, *ceteris paribus*, killing one person to save the lives of two others is the right thing to do.
    c) ... think that actions are right or wrong in themselves.
    d) ... consider happiness the sole good that should guide all of our actions.

17. A decision has to be made between four alternative courses of action.
    On alternative A1, there are three possible outcomes: 20 people dying (probability 0.2), 500 people dying (p = 0.002), or 5 people dying (p = 0.798).
    On alternative A2, there are only two possible outcomes (equal probabilities): 20 people dying, or no one dying.
    On A3, there are three possible outcomes (equal probabilities): 9 people dying, 4 people dying, or 12 people dying.
    On A4, finally, there is a one percent probability of 800 people dying (otherwise, no one dies).
    Select all correct statement(s), given the expected value criterion for decisions involving risks:
    a) A1 is the third best alternative.
    b) A4 is the best alternative.
    c) A3 is the second best alternative.
    d) A2 is the fourth best alternative.

18. Does methodology aim to eliminate all errors in science? Select all correct answers.
    a) No. One argument for this is that science cannot eliminate all errors, only manage potential sources of error.
    b) Yes. Erroneous claims are unscientific.
    c) Yes. Committing errors shows that the related beliefs were unjustified.
    d) No. Available evidence often gives us reason to choose one method over the other but these reasons are often neither complete nor certain.

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19. What is correct about the *hypothetico-deductive* method for hypothesis testing? Select all true claims.

    a) The method determines how hypotheses can be deduced from other, more fundamental theories.
    b) It helps deducing hypotheses from empirical data.
    c) Hypotheses are tested through observable consequences.
    d) If the outcome of a test (i.e. an experiment) is negative, this does not necessarily mean that the hypothesis should be rejected. The negative result might be explained by an auxiliary hypothesis being false.

20. What are necessary conditions for a good operationalisation? Select all correct alternatives.

    a) It must connect the property of interest with an observable property.
    b) The operationalisation must be theory-independent.
    c) The connection between property of interest and observable property must be valid according to our best knowledge.
    d) The observable property must be observable in practice with sufficient precision.

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TaMoS: Answer sheet for multiple choice questions

Student's name: **Correct answers, Natural Science May 31, 2017**

Personal number:

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PART I – Multiple choice questions

Instructions for multiple choice questions: For each question there is *at least* one correct answer alternative. In other words, for each question the number of correct alternatives is 1, 2, 3, or 4. Mark your answers on the separate answer sheet, *not* on the exam. You are awarded one point for each question if, and only if, you have selected the correct alternatives.

1. Does methodology aim to completely eliminate the risk of error in science? Select all correct answers.

   a) No. Errors are a necessary part of accurate claims.
   b) No. Science involves a trade-off between the utility of having knowledge, and the risk of being wrong.
   c) No. Available evidence often gives us reason to choose one method over the other but these reasons are often neither complete nor certain.
   d) No. One argument for this is that science cannot eliminate all errors, only manage potential sources of error.

2. What is the relationship between *scientific methods* and *methodology*? Select all true claims.

   a) For different methods, the same methodology might apply.
   b) Methodology critically assesses the application of methods for certain purposes.
   c) Methods tell you how to do things, methodology justifies why you do things this way rather than that way.
   d) Methodology provides empirical data for problems that methods cannot solve.

3. What is true about *lexical* definitions? Select all true claims.

   a) A lexical definition can be imprecise.
   b) A lexical definition attempts to capture how a dictionary would define a term.
   c) A lexical definition attempts to capture how an expression is normally used.
   d) A lexical definition attempts to capture the common usage of a term.

4. Which of the following sentences satisfy the minimal criteria for hypotheses?

   a) "Do not add water to strong acids!"
   b) "A square is a geometric shape with four sides."
   c) "This teacher gave that student a lower grade because the student was female."
   d) "Individuals with high level of education judge climate change as a high risk."

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5. What distinguishes deductive inferences from inductive inferences? Select all true claims.

   a) Deductions might be true in the right circumstances, inductions are never true.
   b) Valid deduction is truth-preserving, justified induction is fallible.
   c) While justifications of inductive inference rules are controversial, deductive inference rules are "natural" in the sense that they do not require any justification.
   d) Projection is an example of inductive inference, *modus ponens* an example of deductive inferences.

6. What is true of the Duhem-Quine thesis? Select all true claims. According to the Duhem-Quine thesis ...

   a) ... auxiliary hypotheses can be neglected
   b) ... we cannot test our theories claim by claim.
   c) ... falsification is clearly preferable to induction
   d) ... inductive inference are never valid

7. What are *necessary* conditions for a good operationalization? Select all correct alternatives.

   a) The connection between feature of interest and observable property must be valid according to our best knowledge.
   b) The connection between feature of interest and observable property must be sufficiently stable for practical purposes.
   c) The observable property must be observable in practice with sufficient precision.
   d) The feature of interest must be defined with reference to the operationalization.

8. Which of the following claims about systematic error is correct? Select all true claims.

   a) Blinding is a strategy to reduce systematic error.
   b) Increasing construct validity is not a viable strategy to reduce systematic error.
   c) Systematic error can be reduced by repeating measurements and averaging over outcomes.
   d) Systematic errors are (in principle) predictable and deviations from the true value are typically constant or proportional to the true value.

9. When one *ranks* movies on the scale 1,2,3,4,5, what kind of measurement scale is employed? Select all correct alternatives.

   a) A scale that represents less information than an interval scale.
   b) An interval scale.
   c) The kind of scale that is also used when describing the final positions in a marathon.
   d) An ordinal scale.

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10. Which factors would threaten the *internal validity* of inferences from an experiment? Select all correct alternatives.

a) Misreading experimental observations.
b) Failing to accurately identify the relevant background conditions.
c) Misunderstanding the influence of experimental manipulation.
d) Not performing the experiment in a laboratory.

11. Which of the following claims about experiments and observational studies are correct? Select all true claims.

a) Experiments include control and manipulation.
b) Both observational studies and experiments can take place outdoors in a natural environment.
c) In observational studies one does not control all relevant variables, while in experiments one always controls all background factors.
d) Inferences from observational studies might be as valid as from experiments.

12. Which of the following claims about disturbing factors in experiments is correct? Select all true alternatives.

a) The best way to control disturbing factors in an experiment is through statistical techniques.
b) Randomization is an important but not necessary tool for dealing with disturbing factors: by eliminating selection bias and also facilitating blinding.
c) Experiments with disturbing factors always allow externally valid inferences.
d) Observer influence may be a source of disturbing factors in an experiment.

13. Identify the relevant epistemic virtues of scientific models from the following list.

a) Transparency
b) Simplicity
c) Redundancy
d) Non-computability

14. Why should hypotheses be tested statistically? Select all correct options.

a) Statistical accounts often impose weaker (and hence more adequate) demands on hypothesis testing than (non-statistical) accounts of confirmation and falsification.
b) Statistics is the only way to justify inductive inferences.
c) Some hypotheses only make claims about distributions of outcomes.
d) Observations might contain random measurement errors. A single (seemingly) falsifying observation might thus not be a good reason to reject a hypothesis.

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15. Which of these is a way in which Neyman-Pearson Decision Theory and Fisher's Significance Testing are *similar*? Select all correct alternatives.

a) A significant result in the test tells us nothing about the probability of the null hypothesis given the evidence.
b) Hypotheses are only rejected or not - but never accepted as a consequence of the test.
c) Nothing about the probability of the hypotheses is implied by a non-significant result.
d) They take into account the probability of making the observation given that the hypothesis is false.

16. Which of the following are *type* explanations? Select all correct alternatives.

a) The cars crashed because the road went from dual to single lane while one of the cars was trying to overtake.
b) Roads that repeatedly go from double to single lane and back again have higher accident rates because drivers are bad at judging overtaking manoeuvres.
c) Most house fires are due to unattended candles.
d) These five vases broke because they were dropped.

17. What is true of tacit knowledge?

a) Being able to fluently apply the Navier-Stokes equations to complex flow phenomena is an example of tacit knowledge
b) Tacit knowledge is taught mainly through engineering textbooks and manuals
c) The ability to ride a bicycle is an example of tacit knowledge
d) Tacit knowledge is difficult to transfer to another person by means of writing it down or verbalizing it

18. Which claims are true either of methodological individualism or holism? Select all true claims.

a) Methodological holism claims that there are good social explanations that do not provide individualist accounts.
b) Methodological holism defends the view that parts of a whole are in intimate interconnection, such that they cannot exist independently of the whole.
c) Methodological individualism claims that there are no social entities in the world, only individuals (and families).
d) Methodological holism is compatible with the claim that all social phenomena are produced by individual interactions.

19. Which of these are theories of normative ethics? Select all correct answers.

a) Prohibition of plagiarism.
b) Consequentialism.
c) Deontology.
d) Moral justification.

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20. What is true about risk assessment? Select all correct alternatives. Risk assessment is the step in risk analysis where ...

a) ... experiments or observational studies are performed.
b) ... statistical evidence is considered.
c) ... standard scientific techniques are utilized.
d) ... policies are put in place.

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TaMoS: Answer sheet for multiple choice questions

Student's name: **Correct answers, NS versions May 30, 2018**

Personal number: Natural science versions

	a)	b)	c)	d)
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	a)	b)	c)	d)
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1. What is the relationship between *scientific methods* and *methodology*? Select all true claims.

   a) Methodology is the process of constructing specific methods.
   b) Methodology is another word for a collection of methods.
   c) Methodology is about method choice.
   d) Methods tell you how to do things, methodology asks why you do things this way rather than that way.

2. What is true about *lexical* definitions? Select all true claims. A lexical definition is ...

   a) ... a definition intended to capture how the author intends to use the term.
   b) ... a definition that stipulates how a word ought to be used.
   c) ... a definition that captures how a dictionary would define a term.
   d) ... a definition that must be quantified.

3. What is true of Karl Popper's falsificationism? Select all correct answers. The falsificationist...

   a) ... considers only those hypotheses as true that have withstood many falsification attempts.
   b) ... rejects the hypothetico-deductive model.
   c) ... increases her confidence in the truth a hypothesis if this hypothesis is falisfiable.
   d) ... seeks to falsify hypotheses with observable evidence.

4. What is true about Hume's Problem of Induction? Select all true claims. Hume argued ...

   a) ... that induction cannot be justified by foundational principles.
   b) ... that in our inferences, we rely on inference rules that themselves require justification.
   c) ... that inductive inference practices in science should be replaced by deductive inference practices.
   d) ... that the use of inference rules in induction cannot be justified by past successes of applying such a rule.

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5. What distinguishes deductive inferences from inductive inferences? Select all true claims.

   a) Inductions extend conclusions beyond knowledge we already have; deductions order and rearrange our knowledge without adding to its content.
   b) Projection is an example of inductive inference, *modus ponens* an example of deductive inferences.
   c) Valid deduction is truth-preserving, justified induction is fallible.
   d) Generalisation is an example of an inductive inference, *modus tollens* is an example of deductive inferences.

6. What are *necessary* steps in a measurement procedure? Select all correct alternatives.

   a) Specification of a measure and definition of a unit.
   b) Representation by a measurement scale.
   c) Specification of the measure in terms of a physical constant.
   d) Representation by a ratio scale.

7. What are *necessary* conditions for a good operationalization? Select all correct alternatives.

   a) The observable property must be observable in practice with sufficient precision.
   b) The connection between property of interest and observable property must be sufficiently stable for practical purposes.
   c) The operationalization must connect the property of interest with an observable property.
   d) The property of interest must be defined with reference to the operationalization.

8. Which of the following is observable only indirectly? Select all correct alternatives.

   a) The earth's surface.
   b) The precise temperature of a liquid.
   c) The well-being of a child.
   d) Stars inside of our solar system.

9. Which factors would threaten the *internal validity* of inferences from an experiment? Select all correct alternatives.

   a) Misreading experimental observations.
   b) Misunderstanding the influence of experimental manipulation.
   c) Failing to justify why an experimental result applies to a relevant real-world situation.
   d) Failing to accurately identify the relevant background conditions.

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10. What is correct about the *repeatability* of experiments? Select all correct alternatives.

    a) The requirement of repeatability does not mean that scientific experiments actually have to be repeated, only that it should be possible to repeat them given the published description.
    b) An experiment is repeatable if competent persons, different from the experimenters, who follow the given description of the experiment can repeat it.
    c) It is sufficient for an experiment to be repeatable that one of the participating experimenters knows how to repeat the experiment at a later stage.
    d) The criterion of repeatability is satisfied when a person reports that she has done a number of trials with different experimental setups for each trial and out of these trials succeeded in reproducing the same result twice.

11. What is the relation between models and experiments? Select all correct claims.

    a) Experiments and models are similar in that they always are mere representations of targets.
    b) Internal validity is generally less of a problem for models than for experiments.
    c) Both models and experiments involve manipulations. But in models, these are always manipulations of representations.
    d) Models are always representations of targets different from themselves, while experiments are sometimes performed on the target systems themselves.

12. What are good (epistemic or non-epistemic) reasons why scientists build and investigate models of targets they are interested in, rather than investigating these targets directly? Select all correct alternatives.

    a) Science is interested in representations, not in the targets themselves.
    b) We often do not know anything about the target.
    c) Targets can never be manipulated directly.
    d) Ethical or political reasons prohibit the direct manipulation of some targets.

13. Which claims are true about Bayesian statistics? Select all correct alternatives.

    a) In Bayesian statistics, the prior probability of a hypothesis is changed when new evidence is observed.
    b) In Bayesian statistics, no probabilities are assigned to hypotheses.
    c) Bayesian statistics offers an unambiguous algorithm for assigning prior probabilities to hypotheses.
    d) Bayesian statistics can evaluate a potentially large set of competing hypotheses.

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14. Consider the following descriptions of three treatments:
(A) "A medical intervention results in a 33% relative decrease in the incidence of fatal myocardial infarction".
(B) "A medical intervention results in a 2.6 %-point decrease in the incidence of fatal myocardial infarction – 2.7% vs. 5.3%".
(C) "91 persons must be treated with the intervention to reduce the number of fatal myocardial infarction from 3 to 2".
What is true of these descriptions? Select all correct options.

a) It has been shown that most people consider treatments represented in format (C) more effective than treatments represented in format (A).
b) (A),(B) and (C) represent the same risk reduction.
c) Any intervention correctly described by (C) is also correctly described by (A).
d) (C) represents the risk reduction in natural frequencies.

15. What is true of the Deductive-Nomological (DN) Account of Explanation? Select all true answers.

a) According to the DN account, an explanation consists in the inductive inference from initial conditions to the explanandum.
b) According to the DN account, the logical structure of scientific explanations and predictions is very similar.
c) According to the DN account, we achieve genuine understanding of a phenomenon by identifying laws that give reasons to expect the phenomenon to occur.
d) The DN-account cannot use mathematical equations in its explanans.

16. Which claims are true about the relation between correlation and causation? Select all correct claims.

a) Correlation measures the association between variables. Causation measures the productive influence of one variable on another.
b) Mill's method of difference is no longer used in the modern practice of randomized controlled trials.
c) By analyzing correlation data, we typically are able to determine which causal model is the unique explanation of the data.
d) Variables can be correlated with each other even though no direct causal relation exists between them.

17. Which claims are true either of methodological individualism or holism? Select all true claims.

a) Methodological individualism claims that all good social explanations provide individualist accounts of the explanandum.
b) Methodological holism is compatible with the claim that all social phenomena are produced by individual interactions.
c) Methodological holism claims that there are good social explanations that do not provide individualist accounts.
d) Methodological holism defends the view that parts of a whole are in intimate interconnection, such that they cannot exist independently of the whole.

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18. What is true about the *precautionary principle*? Select all correct answers. The precautionary principle ...

a) ... imposes a moral constraint on researchers.
b) ... states a list of research projects that should never be initiated.
c) ... implies that the scientist must be able to show that the new detergent is not harmful, or that the new species does not harm the ecosystem, etc.
d) ... is of no concern in ideal democracies.

19. Which of these are theories of normative ethics? Select all correct answers.

a) Consequentialism.
b) The prohibition of plagiarism.
c) Kantianism.
d) Moral justification.

20. What is true about risk assessment? Select all correct alternatives. Risk assessment is the step in risk analysis where ...

a) ... regulations are determined.
b) ... policies are put in place.
c) ... standard scientific techniques are utilized.
d) ... epidemiological studies are performed.

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TaMoS: Answer sheet for multiple choice questions

Student's name: **Correct answers, Natural Versions October 25, 2017**

Personal number: Natural versions

	a)	b)	c)	d)
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1. What is the relationship between *scientific methods* and *methodology*? Select all true claims.

   a) Methodology is another word for a collection of methods.
   b) Methods tell you how to do things, methodology asks why you do things this way rather than that way.
   c) For different methods, the same methodology might apply.
   d) Methodology is the process of constructing specific methods.

2. What is correct about the peer review system? Select all true claims.

   a) The editor of a scientific journal sends off incoming articles to experts in the area and bases the decision whether to publish or not on their judgment.
   b) It is a system that lets scientists from the same field of research as the author of an article, so called "peer reviewers", critically evaluate the scientific work of that article.
   c) It is a system created to uphold scientific quality of the submitted paper, especially regarding the novelty and justification of its claims.
   d) For any experimental or empirical study, the peer review includes a repetition of the experiment or study.

3. What is true about *inductive inferences*? Select all true claims. Inductive inferences . . .

   a) . . . are fallible.
   b) . . . have premises.
   c) . . . proceed from the general to the particular.
   d) . . . do not guarantee that the conclusions are true.

4. What is correct about the *hypothetico-deductive* method for hypothesis testing? Select all true claims.

   a) The hypothetico-deductive method consists in first testing whether the empirical claim is true and then formulating an hypothesis based on that test.
   b) Hypotheses are tested through observable consequences.
   c) It helps deducing hypotheses from empirical data.
   d) If your hypothesis contains non-directly observable features, their operationalization becomes a necessary part of the hypothetico-deductive method.

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5. When one *ranks* movies on the scale 1,2,3,4,5, what kind of measurement scale is employed? Select all correct alternatives.

   a) The kind of scale that is also used when describing the final positions in a marathon.
   b) An interval scale.
   c) A ratio scale.
   d) A scale that represents more information than an ordinal scale.

6. Which of the following is observable only indirectly? Select all correct alternatives.

   a) The precise temperature of a liquid.
   b) The shape of cells.
   c) The well-being of a child.
   d) Stars outside of our solar system.

7. Which of the following claims about experiments and observational studies are correct? Select all true claims.

   a) In observational studies one does not control all relevant variables, while in experiments one always controls all background factors.
   b) Experiments include control and manipulation.
   c) Both observational studies and experiments can take place outdoors in a natural environment.
   d) Inferences from observational studies might be as valid as from experiments.

8. What is correct about blinding in experiments? Select all correct alternatives.

   a) One condition for double blinding is that experimenters do not know which treatments are administered to what subjects.
   b) Experimental blinding is *not* an instance of experimental control by elimination.
   c) You could use double blinding to reduce observer expectation effects (i.e. interpretation problem).
   d) One condition for single blinding is that subjects do not know which treatment they are getting.

9. What are good (epistemic or non-epistemic) reasons why scientists build and investigate models of targets they are interested in, rather than investigating these targets directly? Select all correct alternatives.

   a) Some targets cannot be handled directly for practical reasons.
   b) Targets can never be manipulated directly.
   c) We often do not know anything about the target.
   d) Some targets are too complex to gain knowledge from their direct investigation.

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10. What distinguishes models from theories? Select all correct claims.

a) Theories allow making true claims, models do not.
b) Theories are empirically supported, models are not.
c) In those scientific domains where models and theories co-exist (e.g. chemistry), theories typically express general principles, which are applied in the building of models.
d) Theories make general claims, models can only represent particulars.

11. Which of the following claims about statistics are true? Select all correct options.

a) It is always clear from the descriptions of the statistical concepts and tools themselves when one should apply them.
b) "Lying with statistics" often consists in the intentional or accidental misapplication of valid statistical methods to good-quality data.
c) Statistics offers different algorithms that might yield different results when applied to the same kind of data.
d) Statistics always lies.

12. Why should hypotheses be tested statistically? Select all correct options.

a) Some hypotheses only make claims about distributions of outcomes.
b) Neither confirmation nor falsification can be based on deductive relationships of hypotheses and observations.
c) Statistics provides tools that turn hypothesis testing into a non-fallible procedure.
d) Observations might contain random measurement errors. A single (seemingly) falsifying observation might thus not be a good reason to reject a hypothesis.

13. Which claims are true about the relation between correlation and causation? Select all correct claims.

a) Mill's method of difference is no longer used in the modern practice of randomized controlled trials.
b) Variables can be correlated with each other even though no direct causal relation exists between them.
c) Correlation measures the association between variables. Causation measures the productive influence of one variable on another.
d) By analyzing merely correlation data, we typically are able to determine which causal model is the unique explanation of the data.

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14. What is true of the Deductive-Nomological (DN) Account of Explanation? Select all true answers.

a) According to the DN account, an explanation consists in the deduction of the explanandum from true generalizations and initial conditions.
b) According to the DN account, an explanation consists in the inductive inference from initial conditions to the explanandum and true generalization.
c) According to the DN account, genuine understanding of a phenomenon is achieved by having reasons for why the phenomenon was expected on a lawful basis.
d) The DN account is problematic, because some cases satisfy the DN account's conditions, but intuitively do not constitute genuine explanations.

15. What are the main claims of anti-naturalist views of the social sciences? Select all correct alternatives.

a) The social sciences, in contrast to the natural sciences, derive law-like generalizations from case studies and qualitative data.
b) The social sciences, in contrast to the natural sciences, are in the business of interpreting human behavior. Interpretation does not produce causal explanations, but some other form of understanding.
c) The social sciences, in contrast to the natural sciences, focus on individual cases and cannot generalize from them.
d) Although the social sciences use qualitative data, in contrast to the natural sciences, there are methods by which these data can be quantified and thus be analyzed with natural science methods.

16. What is true about *informed consent*? Select all correct alternatives. Informed consent ...

a) ... only has support among deontologists who think that individual autonomy is morally important.
b) ... constitutes the legitimate authorization of research participation.
c) ... should never be sought if it risks spoiling the study.
d) ... is usually understood as having three conditions that need to be fulfilled.

17. What is true about risk assessment? Select all correct alternatives. Risk assessment is the step in risk analysis where ...

a) ... experiments or observational studies are performed.
b) ... evidential information is communicated to the public.
c) ... epidemiological studies are performed.
d) ... regulations are determined.

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18. Does methodology aim to completely eliminate the risk of error in science? Select all correct answers.

    a) No. Almost all empirical claims are fallible, so an empirical science that aims to completely eliminate error would lose much of its content.
    b) Yes. Erroneous claims are unscientific.
    c) No. Science involves a trade-off between the utility of having knowledge, and the risk of being wrong.
    d) No. One argument for this is that science cannot eliminate all errors, only manage potential sources of error.

19. What distinguishes valid deductive inferences from inductive inferences? Select all true claims.

    a) Inductions extend conclusions beyond knowledge we already have; deductions order and rearrange our knowledge without adding to its content.
    b) Projection is an example of inductive inference, *modus ponens* an example of deductive inferences.
    c) *Modus tollens* is an example of an inductive inference, projection is an example of a deductive inference.
    d) Deductions might be true in the right circumstances, inductions are never true.

20. Which of the following claims about systematic error is correct? Select all true claims.

    a) Systematic error can be reduced by repeating measurements and averaging over outcomes.
    b) If the cause of a systematic error is identified, then it can usually be corrected.
    c) Increasing construct validity is not a viable strategy to reduce systematic error.
    d) Reducing systematic errors increases accuracy, but not precision.

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TaMoS: Answer sheet for multiple choice questions

Student's name: **Correct answers, Natural Versions August 22, 2017**

Personal number: Natural versions

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19.	◯	◯		
20.		◯		◯

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1. Which of the following claims about words and definitions is/are true? Select all correct alternatives.
   a) A vague word lacks a sharp limit between what it applies to and what it does not apply to.
   b) If a definiens applies to fewer objects than its associated definiendum, then the definition is too broad.
   c) If a definition is too broad, then it cannot be too narrow.
   d) A delimiter typically narrows the scope of what the definiendum applies to.

2. What is the relationship between *scientific methods* and *methodology*? Select all true claims.
   a) Methodology is another word for a collection of methods.
   b) For different methods, the same methodology might apply.
   c) Methods tell you how to do things, methodology justifies why you do things this way rather than that way.
   d) Methodology is the process of constructing specific methods.

3. What is true of the Duhem-Quine thesis? Select all true claims. According to the Duhem-Quine thesis ...
   a) ... we never test a single hypothesis alone.
   b) ... we cannot test our theories claim by claim.
   c) ... falsification is clearly preferable to induction
   d) ... auxiliary hypotheses can be neglected

4. What is true of Karl Popper's falsificationism? Select all correct answers. The falsificationist...
   a) ... never accepts any hypothesis as true – only maintains non-falsified hypotheses as so far not rejected.
   b) ... increases her confidence in the truth of a hypothesis if this hypothesis is falisfiable.
   c) ... considers only those hypotheses as true that have withstood many falsification attempts.
   d) ... rejects the hypothetico-deductive model.

5. What is an ad hoc hypothesis? Select all correct alternatives.
   a) It is a modification of a theory that logically contradicts the rest of the theory.
   b) It is a modification of a theory that attempts to accommodate some unexpected observations in order to save the theory.
   c) It is a modification of a theory that does not in any way explain an observed anomaly.
   d) It is a modification of a theory that decreases falsifiability

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6. What are *necessary* conditions for a good operationalization? Select all correct alternatives.

   a) The operationalization must connect the feature of interest with an observable property.
   b) The connection between feature of interest and an observable property must be valid according to our best knowledge.
   c) The connection between feature of interest and an observable property must be sufficiently stable for practical purposes.
   d) The feature of interest must be defined with reference to the operationalization.

7. Which of the following is true of interval scales? Select all true claims.

   a) One can meaningfully speak of how many times greater or lesser the difference between two points on the scale is than the difference between another pair of points.
   b) Time measured by calendar years (such as ”2015 A.D.” or ”500 BC”) is an interval scale.
   c) Interval scales are identical up to positive linear transformations
   d) If a quantity is measured on an interval scale, it cannot also be measured on a ratio scale.

8. What is correct about blinding in experiments? Select all correct alternatives.

   a) One condition for double blinding is that experimenters do not know which treatments are administered to what subjects.
   b) Blinding can only be realized through randomization.
   c) Single blinding is an effective strategy to reduce placebo effects and thus to increase the internal validity of an experiment.
   d) Double blinding requires that the experimenter is ignorant of which variables are dependent and which are independent.

9. Which of the following claims about disturbing factors in experiments is correct? Select all true alternatives.

   a) When we cannot eliminate disturbing factors, we try to separate them and register their influence independently.
   b) One eliminates a disturbing factor by making sure that the disturbing factor is not present.
   c) Disturbing factors are only present in experiments on humans and animals.
   d) Observer influence may be a source of disturbing factors in an experiment.

10. What are necessary aspects of scientific experiments? Select all correct alternatives.

    a) A theoretical hypothesis.
    b) The laboratory.
    c) The manipulation of some real variable.
    d) The control of background variables.

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11. What distinguishes models from theories? Select all correct claims.

a) In those scientific domains where models and theories co-exist (e.g. chemistry), theories typically express general principles, which are applied in the building of models.
b) Theories make general claims, models can only represent particulars.
c) The main reason to reject a theory is that it is false. The main reason to reject a model is that it is not useful for a given purpose.
d) By proposing a theory, one claims that the theory's object is indeed governed by the principles expressed in the theory. When using a model, one is typically aware of alternative representations of the same object useful for other purposes.

12. What is the relation between models and experiments? Select all correct claims.

a) Performing experimental control is often a difficult task. Setting variable and parameter values in mathematical models, in contrast, is often trivial.
b) Experiments and models are similar in that they always are mere representations of targets.
c) External validity is more of a problem for experiments than for models.
d) Both models and experiments involve manipulations. But in models, these are always manipulations of representations.

13. Which of the following claims about statistics are true? Select all correct options.

a) Since the application of statistical tools necessarily distorts data, statistics always lies.
b) "Lying with statistics" often consists in the intentional misapplication of valid statistical methods to good-quality data.
c) Statistics provides a toolbox of descriptive and inferential methods; and statistical reasoning justifies the choice from this toolbox.
d) The application of valid statistical tools to good-quality data can lead to false conclusions.

14. Consider the following descriptions of three treatments:
(A) "A medical intervention results in a 33% relative decrease in the incidence of fatal myocardial infarction".
(B) "A medical intervention results in a 2.6 %-point decrease in the incidence of fatal myocardial infarction – 2.7% vs. 5.3%".
(C) "91 persons must be treated with the intervention to reduce the number of fatal myocardial infarction from 3 to 2".
What is true of these descriptions? Select all correct options.

a) It has been shown that most people consider treatments represented in format (C) more effective than treatments represented in format (A).
b) Any intervention correctly described by (C) is also correctly described by (A).
c) (A) represents an absolute risk reduction.
d) (B) represents an absolute risk reduction.

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15. What is true about explanans (plural: explanantia) and explanandum (plural: explananda)? Select all true claims.

    a) The explanans is what requires an explanation, the explanandum is what provides the explanation.
    b) Explanantia that only pick out some contributing cause of an explanadum can nevertheless sometimes provide a good explanation.
    c) A singular explanandum can be explained with a singular or general explanans
    d) There are singular and general explananda, but there are only general explanantia.

16. What are the main claims of anti-naturalist views of the social sciences? Select all correct alternatives.

    a) The social sciences must employ their own methods, different from the natural sciences – but both the natural and the social sciences can use the same methodology.
    b) The social sciences, in contrast to the natural sciences, focus on individual cases and cannot generalise from them.
    c) The social sciences cannot and should not follow the methodology of the natural sciences.
    d) The social sciences, in contrast to the natural sciences, are in the business of interpreting human behaviour. Interpretation does not produce causal explanations, but some other form of understanding.

17. Which of the following strategies can help improve the accuracy of interpretation of behavior in a usability experiment? Select all correct alternatives.

    a) Ensure that the belief-desire attributions are circular.
    b) Avoid interpretation of mental states (like beliefs or desires).
    c) Use inter-subjective checks.
    d) Observing indications of cognitive processes, for example a "thinking out loud protocol".

18. What is true about the *precautionary principle*? Select all correct answers. The precautionary principle ...

    a) ... imposes a moral constraint on researchers.
    b) ... states that when an activity raises threats of harm to human health or the environment, precautionary measures should be taken even if some cause and effect relationships are not fully established scientifically.
    c) ... states that the proponent of an activity, rather than the public, should bear the burden of proof of the activity's harmlessness.
    d) ... implies that if scientist A creates a new detergent, then scientist A must be able to show that the new detergent is not harmful.

19. What is true about *informed consent*? Select all correct alternatives. Informed consent ...

    a) ... constitutes the legitimate authorization of research participation.
    b) ... cannot be given involuntarily.
    c) ... is not something that everyone can give.
    d) ... is the faculty dean's approval of the initiation of a new study.

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20. A decision has to be made between four alternative courses of action.
On alternative A, there are two possible outcomes: 100 people dying (probability 0.6) or no one dying (probability = 0.4).
On alternative B, there are four possible outcomes (equal probabilities): 20 people dying, 100 people dying, 400 people dying, or 4 people dying.
On C, there are two possible outcomes: 800 people dying (probability 0.01) or no one dying (probability 0.99).
On D, finally, there are two possible outcomes (equal probabilities): 30 people dying, or no one dying.
Select all correct statement(s), given the expected value criterion for decisions involving risks:

a) B is the best alternative.
b) D is the second best alternative.
c) A is the best alternative.
d) A is the third best alternative.

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TaMoS: Answer sheet for multiple choice questions

Student's name: **Correct answers, NS + CS version January 9 2019**

Personal number: Natural and Computational versions

	a)	b)	c)	d)
1.	○			○
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	a)	b)	c)	d)
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PART I – Multiple choice questions

Instructions for multiple choice questions: For each question there is *at least* one correct answer alternative. In other words, for each question the number of correct alternatives is 1, 2, 3, or 4. Mark your answers on the separate answer sheet, *not* on the exam. You are awarded one point for each question if, and only if, you have selected the correct alternatives and no other. Partial points are not awarded on this part.

1. What is true about knowledge according to the classical definition? Select all correct alternatives. According to the classical definition of knowledge...

   a) ...a person might know P even though she believes P with less than certainty.
   b) ...if a person S knows that it is raining then S believes that it is raining
   c) ...if it is true that it is raining then S knows that it is raining
   d) ...beliefs and justifications are infallible

2. What is true about *lexical* definitions? Select all true claims.

   a) A lexical definition attempts to capture how a dictionary would define a term.
   b) A lexical definition must be quantified.
   c) A lexical definition aims to capture the common usage of a term.
   d) A lexical definition attempts to capture how the author intends to use the term in a particular context.

3. What is true about *falsifiable statements*? Select all true claims. A falsifiable statement...

   a) ...is testable even though it may be about something not directly observable.
   b) ...is neither necessarily true nor necessarily false.
   c) ...is not true.
   d) ...is either true or false.

4. What is correct about the *hypothetico-deductive* method for hypothesis testing? Select all true claims.

   a) Hypotheses are tested through observable consequences.
   b) It is a procedure by which one deduces a testable hypothesis from empirical data.
   c) The hypothetico-deductive method allows us to deductively confirm hypotheses.
   d) If your hypothesis contains non-directly observable features, the operationalization of these features becomes a necessary part of the hypothetico-deductive method.

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5. What are *necessary* conditions for a good operationalization? Select all correct alternatives.

   a) One should only operationalize hypotheses that satisfy the quality criteria for hypotheses.
   b) The connection between feature of interest and a directly observable property must be valid according to our best knowledge.
   c) The connection between feature of interest and a directly observable property must be sufficiently stable for practical purposes.
   d) The observable property must be observable in practice with sufficient precision.

6. Which of the following claims about systematic error is correct? Select all true claims.

   a) Experimental blinding is a strategy to reduce systematic error.
   b) Increasing construct validity is not a viable strategy to reduce systematic error.
   c) If the cause of a systematic error is identified, then the error can usually be corrected.
   d) Systematic error can be reduced by repeating measurements and averaging over outcomes.

7. What is true about Mill's method of difference?

   a) It requires successful control and intervention.
   b) It requires statistical techniques.
   c) By activating the same factor in two separate, but identical, systems we can determine the effect of that factor.
   d) According to Mill's method of difference, we can determine causal relations by performing experiments.

8. Which of the following claims about disturbing factors in experiments is correct? Select all true alternatives.

   a) When we cannot eliminate disturbing factors, we can try to separate them and register their influence independently
   b) Disturbing factors are only present in experiments on humans and animals.
   c) Observer influence can be a disturbing factor in an experiment.
   d) Experiments with disturbing factors typically allow for externally valid inferences.

9. What is true regarding the virtues of models?

   a) A model is simple if changing one of the assumptions in the model does not change the model result.
   b) Theoretical tractability means that a model can be processed by a theory in a scientifically useful way.
   c) If changing one of the assumptions in a model does not change the model result, then the model is robust with respect to this assumption.
   d) When comparing two models, a model A is more precise than another model B if A's parameter specifications imply B's parameter specifications.

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10. What are good (epistemic or non-epistemic) reasons why scientists build and investigate models of targets they are interested in, rather than investigating these targets directly? Select all correct alternatives.

a) It is too expensive to handle some targets directly.
b) Targets can never be manipulated directly.
c) We often do not know anything about the target, and modelling does not require target information.
d) Science is interested in representations, not in the targets themselves.

11. Which claims are true of Fisher's Significance Testing? Select all correct options.

a) For deterministic hypothesis, the relevant test statistic for Fisher's Significance Test is determined by information about random error in the experiment.
b) The p-value is independent of the tested hypothesis.
c) The result of Fisher's Significance Testing can be influenced by changing the experimental design, and by changing the partition of possible outcomes.
d) Fisher's Significance Test only helps decide whether to reject the hypothesis or not.

12. Which claims are true about Bayesian statistics? Select all correct alternatives.

a) Bayesian statistics offers an unambiguous algorithm for assigning prior probabilities to hypotheses.
b) In Bayesian statistics, the probability of a hypothesis is changed if new evidence is observed.
c) Bayesian statistics can evaluate a large set of competing hypotheses.
d) Bayesian statistics interprets probabilities as "degrees of belief" or "subjective confidence in hypotheses".

13. Which claims are true about the relation between correlation and causation? Select all correct claims.

a) Correlation measures the association between variables. Causation measures the productive influence of one variable on another.
b) Mill's method of difference is no longer used in the modern practice of randomized controlled trials.
c) Two variables can be correlated with each other even though neither one is the cause of the other.
d) Correlation between two variables is symmetric. The causal relation between two variables in asymmetric.

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14. What is true regarding the structure of explanations?

a) Type explanations can provide general explanations to singular phenomena, i.e. the explanans are general principles.
b) A singular explanation explains a particular event with references to the relevant aspects of the particular situation.
c) An explanation features an explanans, something to be explained, and an explanandum, something which provides the explanation.
d) An explanation features an explanandum, something to be explained, and an explanans, something which provides the explanation.

15. Engineering is not just applied science. Why is that? Mark all correct answers.

a) Engineering and science are fundamentally disconnected.
b) Artefact design and functional analysis are two examples of non-scientific methods used by engineers.
c) You cannot always apply the correct scientific theory to an engineering problem.
d) Engineers are often unaware of what the relevant scientific theory is that should be applied.

16. What is true about belief-desire explanations? Select all correct alternatives. Belief-desire explanations ...

a) ... cannot be improved because they depend on subjective interpretations.
b) ... explain human behavior as a result of mental states.
c) ... are often used to explain human behavior.
d) ... can be used to explain how aims and available resources of individuals influence their behavior.

17. In the social sciences, how can the quality of an interpretation of human action be improved?

a) Only rely on mental states that are directly observable.
b) Use intersubjective checks of the interpretation, e.g. by involving multiple researchers in the analysis.
c) Check for coherence between answers, e.g. through (slightly reformulated) repeat questions.
d) Rely exclusively on a robust correlation between stimulus inputs and behavioral outputs.

18. What is the relation between models and experiments? Select all correct claims.

a) Internal validity is generally less of a problem for models than for experiments.
b) Experiments and models are similar in that they always are mere representations of targets.
c) Both models and experiments involve manipulations. But in models, these are manipulations of representations.
d) External validity is more of a problem for experiments than for models.

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19. What is true about the *precautionary principle*? Select all correct answers. The precautionary principle ...

a) ... states that when experiments exceed the research budget, they should not be conducted.
b) ... states that the proponent of an activity, rather than the public, should bear the burden of proof of the activity's harmlessness.
c) ... imposes a moral constraint on researchers.
d) ... states a list of research projects that should never be initiated.

20. What is true about *deontology*? Deontologists ...

a) ... think that actions are right or wrong in themselves.
b) ... consider happiness the sole good that should guide all of our actions.
c) ... tend to hold duties or rights as more important than states of affairs for moral evaluation.
d) ... are more inclined than others to subscribe to the validity of categorical imperatives.

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TaMoS: Answer sheet for multiple choice questions

Student's name: **Correct answers, NS version April 16 2019**

Personal number: Natural versions

	a)	b)	c)	d)
1.	◯	◯		
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PART I – Multiple choice questions

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1. What is the relationship between *scientific methods* and *methodology*? Select all true claims.

   a) Methodology is another word for constructing specific methods.
   b) Methods tell you how to do things, methodology justifies why you do things this way rather than that way.
   c) For different methods, the same methodology might apply.
   d) Methodology is about method choice.

2. What is true about the justification of method choice?

   a) Satisfying epistemic goals is not the prime justification of method choice.
   b) The practice of justifying method choice is called methodology.
   c) Scientists should choose the methods which give the best economic results.
   d) All ways of justifying method choice satisfy the same goals.

3. Which of the following sentences satisfies the minimal criteria for hypotheses?

   a) "A square is a geometric shape with four sides."
   b) "Is Amyloid plaque the cause of Alzheimer's?"
   c) "Do not add water to strong acids!"
   d) "Computer program of type A uses more memory than than computer program of type B."

4. What is an ad hoc hypothesis? Select all correct alternatives.

   a) It is a modification of a theory that does not in any way explain an observed anomaly.
   b) It is a modification of a theory that attempts to accommodate some unexpected observations only in order to save the theory.
   c) It is a modification of a theory that decreases falsifiability.
   d) It is a modification of a theory that does not propose a new means of testing the theory.

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5. What distinguishes deductive inferences from inductive inferences? Select all true claims.

   a) Deductions might be true in the right circumstances, inductions are never true.
   b) While justifications of inductive inference rules are controversial, deductive inference rules are "natural" in the sense that they do not require any justification.
   c) Inductions extend conclusions beyond knowledge we already have; deductions order and rearrange our knowledge without adding to its content.
   d) Modus tollens is an example of an inductive inference, projection is an example of a deductive inference.

6. Which of the following is true of interval scales? Select all true claims.

   a) Interval scales are identical up to positive linear transformations
   b) If a quantity is measured on an interval scale, it cannot also be measured on a ratio scale.
   c) Zero on the scale represents the complete absence of the property concerned.
   d) On (linear) interval scales, arithmetic means are meaningful as averages.

7. When one *ranks* movies on the scale 1,2,3,4,5, what kind of measurement scale is employed? Select all correct alternatives.

   a) A scale that represents less information than an interval scale.
   b) An interval scale.
   c) A scale that represents more information than an ordinal scale.
   d) An ordinal scale.

8. What is true about Randomised Control Trials (RCTs)?

   a) RCTs facilitates both single and double blinding.
   b) It guarantees that background factors are equally distributed between test and control in a test.
   c) Using RCT implies that you are using double blinding.
   d) It does not guarantee that background factors are equally distributed between test and control in a test.

9. Which factors would threaten the *internal validity* of inferences from an experiment? Select all correct alternatives.

   a) Misreading experimental observations.
   b) Failing to justify why an experimental result applies to a relevant real-world situation.
   c) Failing to accurately identify the relevant background conditions.
   d) If randomization was used in a field experiment.

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10. What distinguishes models from theories? Select all correct claims.
    a) By proposing a theory, one claims that the theory's target is indeed governed by the principles expressed in the theory. When using a model, one is typically aware of alternative representations useful for other purposes.
    b) Theories allow making true claims, models do not.
    c) Theories make general claims, models can only represent particulars.
    d) Theories are empirically supported, models are not.

11. What is true about valid isolating models? Select all correct claims.
    a) A valid isolating model represents the workings of the isolated factor accurately.
    b) A valid isolating model singles out one or a few aspects of the target, ignoring all the others.
    c) Isolating models can never be validated by experiments.
    d) A valid isolating model is is similar to all all of the target's features.

12. Which claims are true about Bayesian statistics? Select all correct alternatives.
    a) Bayesian statistics interprets probabilities as "degrees of belief" or "subjective confidence in hypotheses".
    b) In Bayesian statistics, the probability of a hypothesis is changed as a consequence of observing new evidence.
    c) The main goal in Bayesian hypothesis evaluation is to decide whether to accept a single hypothesis as either true or false.
    d) Bayesian statistics offers an unambiguous algorithm for assigning prior probabilities to hypotheses.

13. Which of these is a way in which Neyman-Pearson Decision Theory and Fisher's Hypothesis Testing are *similar*? Select all correct alternatives.
    a) More than one hypothesis can be tested simultaneously.
    b) Hypotheses are only rejected or not - but never accepted as a consequence of the test.
    c) The probability for the outcome given a hypothesis is compared to a threshold level.
    d) Observations might contain random measurement errors. A single (seemingly) falsifying observation might thus not be a good reason to reject a hypothesis.

14. Which claims are true about the relation between correlation and causation? Select all correct claims.
    a) Correlation between two variables is symmetric. The causal relation between two variables in asymmetric.
    b) Many different causal models between two variables are compatible with the observation that the two variables are correlated.
    c) Two variables can be correlated with each other even though neither one is caused by the other.
    d) That there is a common cause of two variables X and Y explains why there is a causal relation between X and Y (either X causes Y or Y causes X), although there is no correlation between X and Y.

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15. Which of the following are *type* explanations? Select all correct alternatives.

a) The cars crashed because the road went from dual to single lane while one of the cars was trying to overtake.
b) Birds are able to follow their migration routes because they have receptors in their eyes that are sensitive to the magnetic field of the earth.
c) These five vases broke because they were dropped.
d) The house burned down because a candle was left to burn unattended.

16. What is true of tacit knowledge?

a) Tacit knowledge is taught mainly through engineering textbooks and manuals.
b) Transferring tacit knowledge typically requires personal contact and interaction.
c) To know something tacitly means that one has the ability to describe this knowledge intuitively, i.e. without first having to study it.
d) The ability to ride a bicycle is an example of tacit knowledge.

17. What are the main claims of anti-naturalist views of the social sciences? Select all correct alternatives.

a) The social sciences cannot and should not follow the methodology of the natural sciences.
b) The social sciences, in contrast to the natural sciences, focus on individual cases and cannot generalise from them.
c) Although the social sciences use qualitative data, in contrast to the natural sciences, there are methods by which these data can be quantified and thus be analyzed with natural science methods.
d) The social sciences are fundamentally different from the natural sciences, for instance because the social sciences work with qualitative data and case studies which the natural sciences do not.

18. What is true about belief-desire explanations? Select all correct alternatives. Belief-desire explanations ...

a) ... can be used to explain how aims of individuals influence their behavior.
b) ... explain human behavior as a result of mental states.
c) ... entail that the ends always justify the means.
d) ... are not causal explanations.

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19. A decision has to be made between four alternative courses of action.
On alternative A, there are two possible outcomes: 100 people dying (probability 0.6) or no one dying (probability = 0.4).
On alternative B, there are four possible outcomes (equal probabilities): 20 people dying, 100 people dying, 400 people dying, or 4 people dying.
On C, there are two possible outcomes: 800 people dying (probability 0.01) or no one dying (probability 0.99).
On D, finally, there are two possible outcomes (equal probabilities): 30 people dying, or no one dying.
Select all correct statement(s), given the expected value criterion for decisions involving risks:

a) C is the second best alternative.
b) C is the best alternative.
c) A is the third best alternative.
d) A is the best alternative.

20. What is true about the *precautionary principle*? Select all correct answers.
The precautionary principle ...

a) ... states that when experiments exceed the research budget, they should not be conducted.
b) ... is of no concern in ideal democracies.
c) ... implies that if scientist A creates a new detergent, then scientist A must be able to show that the new detergent is not harmful.
d) ... states that if an action or policy has a suspected risk of causing harm to the public or to the environment, in the absence of scientific consensus that the action or policy is harmful, the burden of proof that it is not harmful falls on those taking an action.

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TaMoS: Answer sheet for multiple choice questions

Student's name: **Correct answers, May 29, 2019, nat. sci. version**

Personal number:

	a)	b)	c)	d)
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PART I – Multiple choice questions

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1. A part of the classical definition of knowledge concerns belief. What is true regarding the concept of belief?

   a) It is possible to believe something that is not true.
   b) Beliefs are properties of scientific theories.
   c) Beliefs are action guiding.
   d) Beliefs do not come in degrees.

2. What is the relationship between *scientific methods* and *methodology*? Select all true claims.

   a) Methodology concerns method choice.
   b) Methodology is another word for a collection of methods.
   c) For different methods, the same methodology might apply.
   d) Methodology provides empirical data for problems that methods cannot solve.

3. What is true of Karl Popper's falsificationism? Select all correct answers. The falsificationist...

   a) ...seeks to falsify hypotheses with observable evidence.
   b) ...considers only those hypotheses as true that have withstood many falsification attempts.
   c) ...rejects the hypothetico-deductive model.
   d) ...never accepts hypotheses as true – only maintains non-falsified hypotheses as so far not rejected.

4. Which of the following sentences can be hypotheses?

   a) "Remember to clean the equipment between trials!"
   b) "Asphalt of type C1 has a longer life-span than type C2."
   c) "Mercury thermometers do not give accurate results at temperatures close to -37 degrees Celsius."
   d) "People in rural areas have a lower tolerance to viruses."

5. What is an *ad hoc* hypothesis? Select all correct alternatives.

   a) It is a modification of a theory that *increases* falsifiability.
   b) It is a modification of a theory that *decreases* falsifiability.
   c) It is a modification of a theory that does not explain an observed anomaly.
   d) It is a modification of a theory that attempts to accommodate some unexpected observations only in order to save the theory.

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6. What is true regarding measurement error?

   a) If a measurement unit is based on a prototype that changes over time, the measurement's stability will be threatened.
   b) The stability of the measurement is mainly threatened by random error.
   c) If a measurement is made several times and gives similar values, then the measurement is precise.
   d) A measurement can be both inaccurate and imprecise at the same time — if the operationalization is affected by both systematic and random error.

7. Which of the following is true of interval scales? Select all true claims.

   a) On (linear) interval scales, arithmetic means are meaningful as averages.
   b) Interval scales are identical up to positive linear transformations.
   c) If a quantity is measured on an interval scale, it cannot also be measured on a ratio scale.
   d) 30 degrees Celsius is twice as warm as 15 degrees Celsius.

8. Which of the following claims about experiments and observational studies are correct? Select all true claims.

   a) An inferences from an observational study might be as justified as an inference from an experiment.
   b) Both observational studies and experiments can take place outdoors in a natural environment.
   c) Observational studies do not include an intervention.
   d) It is a requirement for something to be an experiment that one successfully controls all relevant variables.

9. What are necessary aspects of scientific experiments? Select all correct alternatives.

   a) A statistical model.
   b) An intervention.
   c) A hypothesis derived from a scientific theory.
   d) A manipulation of some real variable.

10. Which of the following properties are epistemic virtues of scientific models?

    a) Simplicity.
    b) Precision.
    c) Robustness.
    d) Non-computability.

11. What is the relation between models and experiments? Select all correct claims.

    a) External validity is more of a problem for experiments than for models.
    b) Experiments and models are similar in that they always are mere representations of targets.
    c) Both models and experiments involve manipulations. But in models, these are always manipulations of representations.
    d) Internal validity is generally less of a problem for models than for experiments.

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12. In what ways are Neyman-Pearson Decision Theory and Fisher's Hypothesis Testing *similar*? Select all correct alternatives.

    a) The tests always result in the acceptance of a hypothesis.
    b) The tests say nothing about the probability of the hypothesis.
    c) More than one hypothesis can be tested simultaneously.
    d) The probability of the outcome given a hypothesis is compared to a threshold level.

13. Which of the following claims about statistics are true? Select all correct options.

    a) Statistics always lies.
    b) "Lying with statistics" consists in the misapplication of valid statistical methods to good-quality data.
    c) It is always clear from the descriptions of the statistical concepts and tools themselves when one should apply them.
    d) One can only "lie with statistics" by using invalid statistical methods, or by applying valid statistical methods to junk data.

14. Which claims are true about the relation between correlation and causation? Select all correct claims.

    a) Many different causal models of two variables are compatible with the observation that the two variables are correlated.
    b) Two variables can be correlated with each other even though neither one is caused by the other.
    c) A correlation between two variables is symmetric. A causal relation between two variables is asymmetric.
    d) That two variables are uncorrlated means that there is no causal relation between them.

15. What is true regarding the two theories of justification?

    a) Foundationalism suggests that a claim is justified only if it is inferred from a set of self-evident axioms.
    b) Hume's problem of induction is not a problem for a foundationalist.
    c) Hume's problem of induction is not a problem for a coherentist.
    d) Coherentism favour justification consisting of a coherent system with accepted claim, and any claim is justified if the claim can be incorporated in the system.

16. What is true regarding the relationship between "design" and "science"?

    a) Design, unlike science, generally does not provide any sort of justification for method choice.
    b) Science is guided by norms such as efficiency, sustainability and user-friendliness, which design is not guided by.
    c) The goals of science are explanation and prediction, whereas the goal of design is the satisfaction of functions.
    d) The common aim of science and design is to both describe and change the world.

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17. In what ways does qualitative and quantitative data differ?

    a) Quantitative measurements require operationalisations; qualitative measurements do not.
    b) Qualitative data contains information about a wide variety of aspects, while quantitative data typically contains information only about a few properties.
    c) Large qualitative data sets are more difficult to summarize than large quantitative data sets.
    d) Quantitative data can be valid, whereas qualitative data cannot be valid or invalid, but should be evaluated as ”authentic” or ”inauthentic” instead.

18. Which of the following strategies can help improve the accuracy of interpretation of behavior in a usability experiment? Select all correct alternatives.

    a) Check for coherence by repeating questions in questionnaire or interview.
    b) Make ad hoc modifications whenever suitable.
    c) Ensure that the belief-desire attributions are circular.
    d) Observing indications of cognitive processes, for example a ”thinking out loud protocol”.

19. Which of these are correct description rules used to manage uncertainty whose probabilities cannot be quantified?

    a) "MaxiMax": We should identify the option with the worst outcome and choose that option.
    b) "MaxiMin": We should identify the option with the least bad outcome and choose that option.
    c) "The principle of sufficient reason": we should assume the current state of knowledge is enough to determine the best option.
    d) "The precautionary principle": the proponent of a possibly harmful activity bears the burden of proving that it is harmless.

20. Which of these are theories of normative ethics? Select all correct answers.

    a) Virtue ethics.
    b) Consequentialism.
    c) Deontology.
    d) Informed consent.

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TaMoS: Answer sheet for multiple choice questions

Student's name: **Correct answers, October 23, 2019, Nat sci. version**

Personal number:

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Option 5c will not be considered. One plausible reading of the sentence made the option correct.

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PART I – Multiple choice questions

Instructions for multiple choice questions: For each question there is *at least* one correct answer alternative. In other words, for each question the number of correct alternatives is 1, 2, 3, or 4. Mark your answers with "X" on the separate answer sheet, *not* on the exam. You are awarded one point for each question if, and only if, you have selected the correct alternatives and no other. Partial points are not awarded on this part.

1. What is *realism* in the theory of science? Select all true claims. Realism is...

   a) ...for example the view that electrons exist, even though they are not directly observable.
   b) ...the view that scientific theories describe, or aim at describing, what the world is really like.
   c) ...the view that scientific concepts are not descriptions of the actual world, but merely convenient fictions with which scientists can construct coherent theories.
   d) ...the view that we have reason to believe that our best theories are literally true.

2. What is true about *lexical* definitions? Select all true claims.

   a) A lexical definition aims to capture the common usage of a term.
   b) A lexical definition must be quantified.
   c) A good lexical definition is neither too broad nor too narrow.
   d) A lexical definition stipulates how a word ought to be used.

3. What is true regarding the problems with falsificationism and the hypothetico-deductive method?

   a) The Duhem/Quine thesis poses potential problems for falsificationism.
   b) According to falsificationism hypotheses which one believes to be true are assigned a higher confidence.
   c) In actual scientific practice, there is an asymmetry between confirmation and falsification.
   d) In actual scientific practice, hypotheses have confidences, some are considered more likely to be true.

4. What is true regarding the two kinds of justification?

   a) Hume's problem of induction is not a problem for a foundationalist.
   b) Hume's problem of induction is not a problem for a coherentist.
   c) Foundationalism provides justification by identifying basic claims from which the claims to be justified can be inferred.
   d) Coherentism provides justification consisting of a non-contradictory system with accepted claims, and any claim is justified if the claim can be incorporated in the system.

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5. What is true about *falsifiable statements*? Select all true claims. A falsifiable statement...

   a) ...is necessarily false.
   b) ...is always a claim about a particular observable phenomenon and never about a general trend or theory.
   c) ...is not true.
   d) ...is testable even though it may be about something not directly observable.

6. What are *necessary* conditions for a good operationalization? Select all correct alternatives.

   a) The feature of interest must be defined with reference to the operationalization.
   b) The connection between a feature of interest and a directly observable property must be sufficiently stable for practical purposes.
   c) The observable property must be observable in practice with sufficient precision.
   d) The operationalization must connect the feature of interest with an observable property.

7. What is true regarding theory-dependence?

   a) Some observations can be made without relying on theory.
   b) Since scientists gather data, from which they infer theory, basic research is thus not theory-dependent.
   c) Aided observations are not theory-dependent.
   d) Some theories help making observations, which in turn justify these theories — this is called the experimenter's regress.

8. What is correct about blinding in experiments? Select all correct alternatives.

   a) Experimental blinding is *not* an instance of experimental control by elimination.
   b) One condition for double blinding is that experimenters do not know which treatments are administered to what subjects.
   c) Single blinding is an effective strategy to reduce the impact of placebo effects and thus to increase the internal validity of an experiment.
   d) One condition for single blinding is that subjects do not know if they are receiving treatment or not.

9. What are necessary aspects of scientific experiments? Select all correct alternatives.

   a) A statistical model.
   b) A laboratory.
   c) An observation of the result of manipulation.
   d) A manipulation of some real variable.

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10. What distinguishes models from theories? Select all correct claims.

a) Theories make general claims, models can only represent particulars.
b) Scientists who use a theory have reasons to believe that it is true, while scientists who use a model typically know that it is false in several respects.
c) Theories are empirically supported, models are not.
d) By proposing a theory, one claims that the theory's target is indeed governed by the principles expressed in the theory. When using a model, one is typically aware of alternative representations that are useful for other purposes.

11. What is true regarding the virtues of models?

a) When comparing two models, the one which implies the parameter specifications of the other, is more robust with respect to that parameter.
b) When a model user can understand how the model produces the result, then it is epistemically transparent.
c) A model is simple if changing one of the assumptions in the model does not change the model result.
d) Transparency means that a model follows some scientific theory.

12. Consider the following descriptions of three treatments:
(A) "A medical intervention results in a 33% relative decrease in the incidence of fatal myocardial infarction".
(B) "A medical intervention results in a 2.6 %-point decrease in the incidence of fatal myocardial infarction – from 5.3%' to 2.7% ".
(C) "91 persons must be treated with the intervention to reduce the number of fatal myocardial infarction from 3 to 2".
What is true of these descriptions? Select all correct options.

a) (A), (B) and (C) represent the same risk reduction.
b) (C) represents the risk reduction in natural frequencies.
c) Any intervention correctly described by (C) is also correctly described by (A).
d) (B) represents an absolute risk reduction.

13. Which claims are true about Bayesian statistics? Select all correct alternatives.

a) Bayesian statistics offers an unambiguous algorithm for assigning prior probabilities to hypotheses.
b) The "problem of old evidence" remains a substantial problem for Bayesian statistics.
c) The main goal in Bayesian hypothesis evaluation is to decide whether to accept a single hypothesis as either true or false.
d) Bayesian statistics interprets probabilities as "degrees of belief" or "subjective confidence in hypotheses".

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14. What is true regarding explanatory virtues (what makes explanations powerful)?

    a) An explanation is more powerful than another, if it is less sensitive to background causes.
    b) A low level of detail in the description of the phenomenon to be explained is synonymous with having precision of the explanandum.
    c) An explanation which as its explanans provides correct complex and intricate information is often considered to be cognitively salient.
    d) An explanans should identify those contributing causes that produce the difference asked for in the contrast of the explanandum.

15. Which claims are true about the relation between correlation and causation? Select all correct claims.

    a) Correlation between two variables is a symmetric relation. The causal relation between two variables is asymmetric.
    b) Many different causal models of two variables are compatible with the observation that the two variables are correlated.
    c) Two variables can be correlated with each other even though neither one is caused by the other.
    d) From data showing that two variables are correlated, we are typically able to determine which causal model is the unique explanation of the data.

16. What is true regarding the steps taken in the methodology of design?

    a) The first step is to perform an exhaustive search for a complete set of design alternatives.
    b) The design needs to be validated, which means to identify a possible physical description.
    c) The design needs to be validated, which means to show that it actually fulfils the functional requirements.
    d) An important step is identifying possible physical descriptions, for instance by considering operational principles and normal configuration.

17. What are the main claims of anti-naturalist views of the social sciences? Select all correct alternatives.

    a) The social sciences are fundamentally different from the natural sciences, for instance because the social sciences work with qualitative data and case studies which the natural sciences do not.
    b) The social sciences cannot and should not follow the methodology of the natural sciences.
    c) The social sciences must employ their own methods, different from the natural sciences – but both the natural and the social sciences can use the same methodology.
    d) The social sciences derive law-like generalizations from case studies and qualitative data.

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18. In what ways do qualitative and quantitative data differ?

a) Large qualitative data sets are more difficult to summarize than large quantitative data sets.
b) Qualitative and quantitative data have no information in common - they have fundamentally different content.
c) Qualitative data contain information about a wide variety of aspects, while quantitative data typically contain information only about a few properties.
d) Scientists often use different methods to produce qualitative and quantitative data, respectively.

19. A decision has to be made between four alternative courses of action.
On alternative A, there are two possible outcomes: 100 people dying (probability 0.6) or no one dying (probability = 0.4).
On alternative B, there are four possible outcomes (equal probabilities): 20 people dying, 100 people dying, 400 people dying, or 4 people dying.
On C, there are two possible outcomes: 800 people dying (probability 0.01) or no one dying (probability 0.99).
On D, finally, there are two possible outcomes (equal probabilities): 30 people dying, or no one dying.
Select all correct statement(s), given the expected value criterion for decisions involving risks:

a) C is the best alternative.
b) B is the fourth best alternative.
c) C is the second best alternative.
d) B is the best alternative.

20. What is true about the *precautionary principle*? Select all correct answers.
The precautionary principle ...

a) ...states that if an action or policy has a suspected risk of causing harm to the public or to the environment, in the absence of scientific consensus that the action or policy is harmful, the burden of proof that it is not harmful falls on those taking an action.
b) ...states that when experiments exceed the research budget, they should not be conducted.
c) ...states a list of research projects that should never be initiated.
d) ...is of no concern in ideal democracies.

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TaMoS: Answer sheet for multiple choice questions

Student's name: **Correct answers, December 16, 2019, Nat sci. version**

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